AI Agent for Coding
Code Smarter, Not Harder

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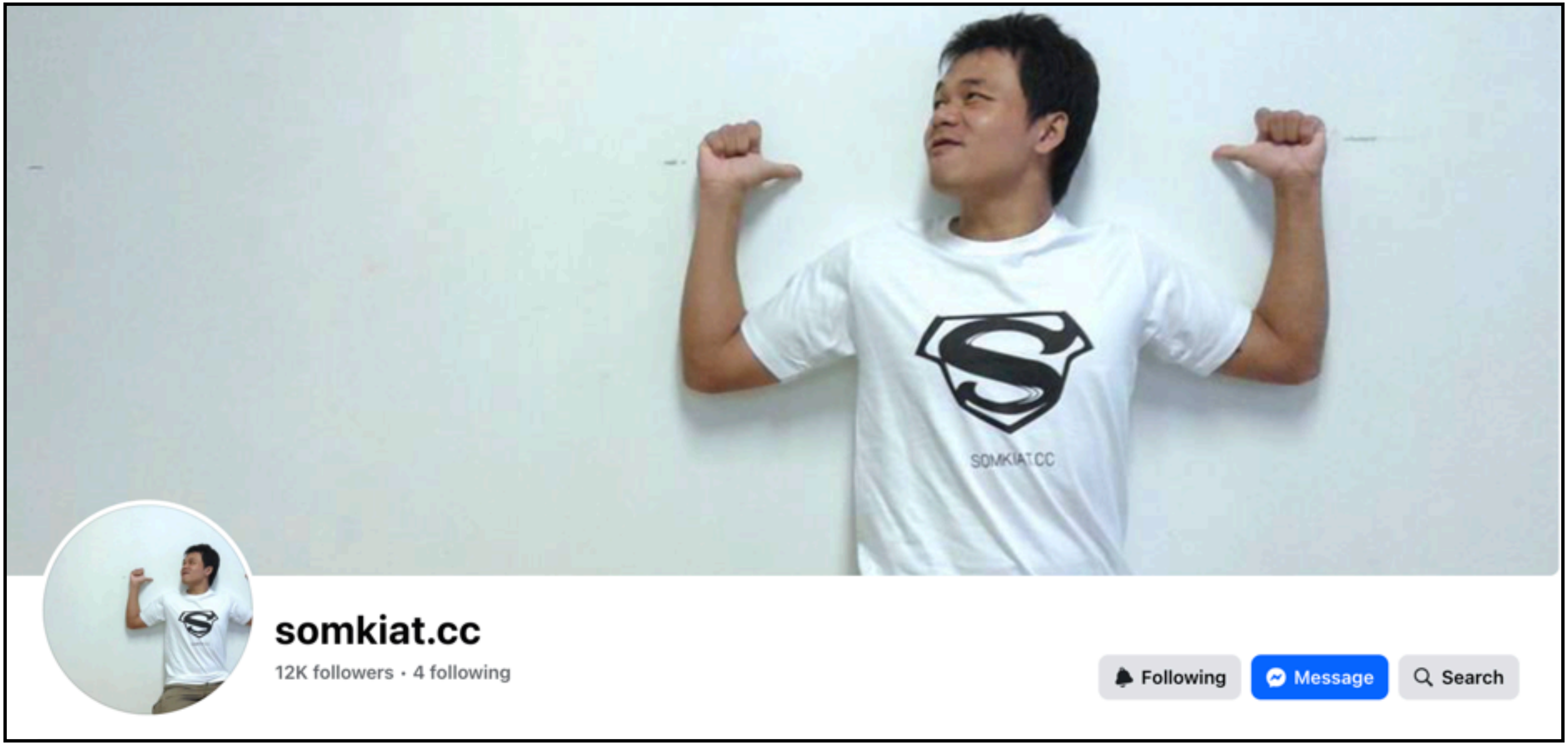

Software Developer at SCK

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Slide นี้ คนทำนะครับ ไม่ได้ใช้ Gen AI !!

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AI Agent for Coding
Code smarter, Not harder

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Topics

Software development workflows with AI Agent
Spec-Driven Development (SDD)
Demo coding with AI Agent

Requirement

Planning

Coding
Testing

Validate and Testing

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Experiment > Optimize

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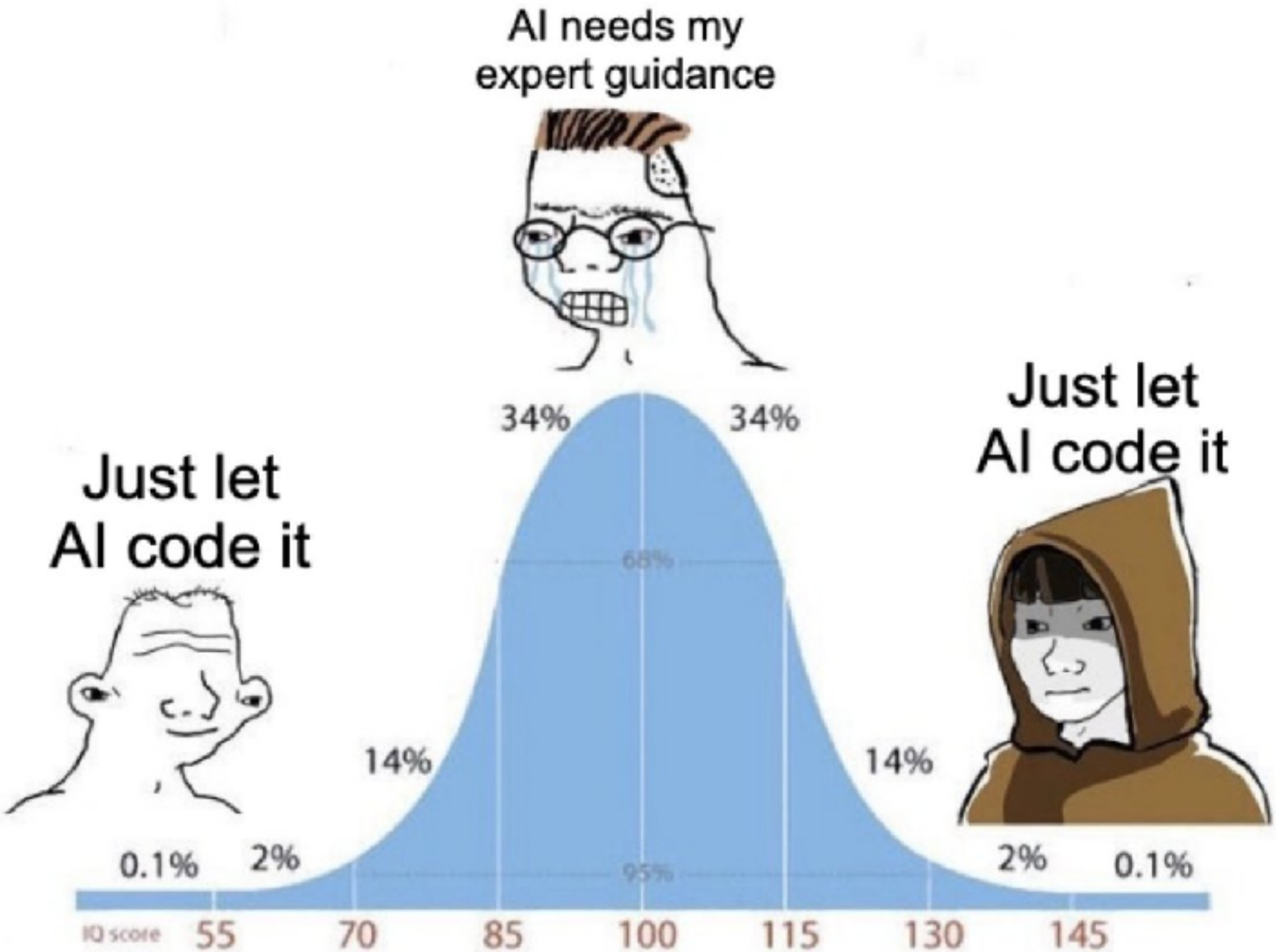

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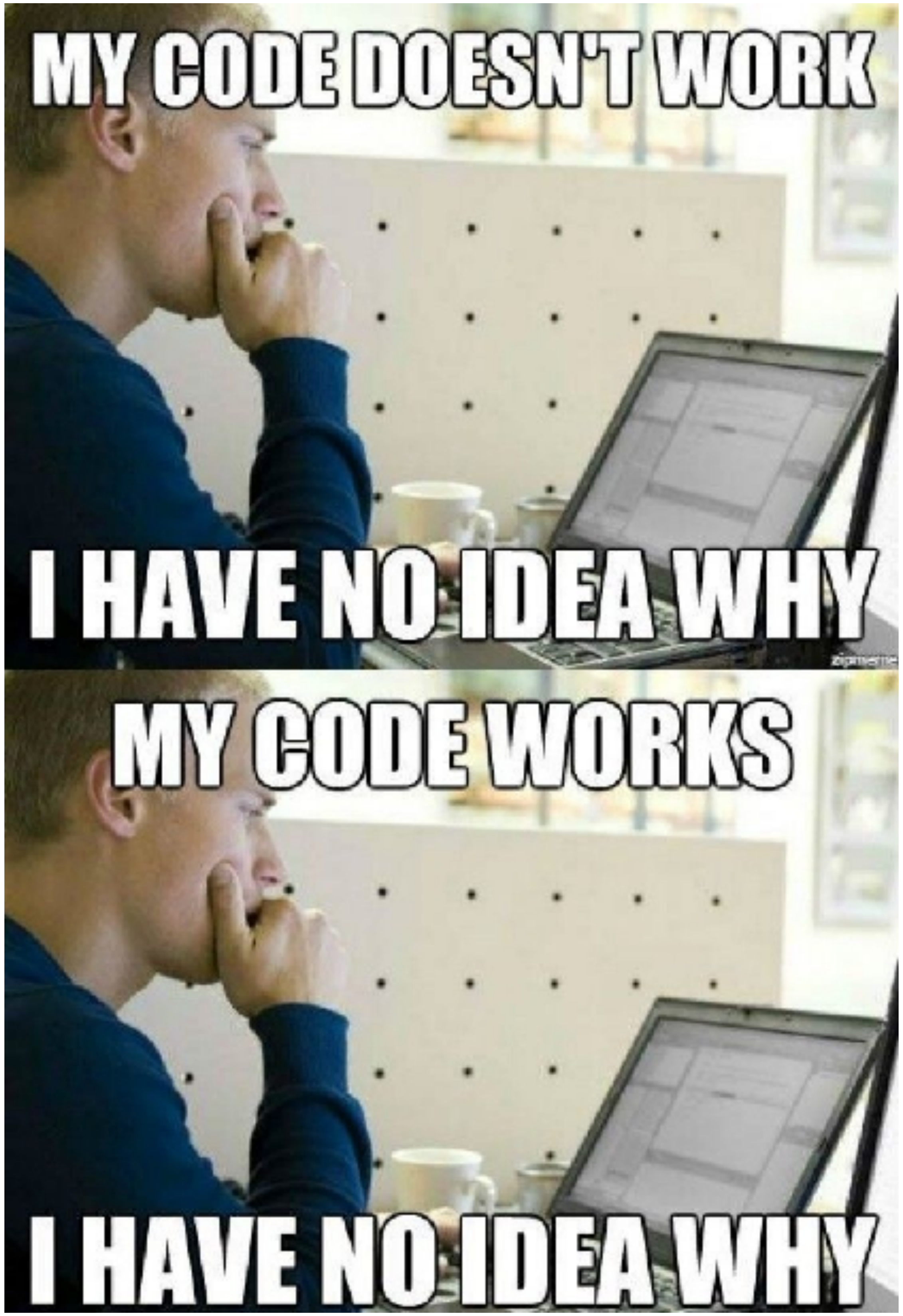

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History of Programming

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History of Programming

Machine code → Assembly → High-level language → AI-assist coding → Vibe coding

Increase of abstraction

Productivity

Knowledge gap !!

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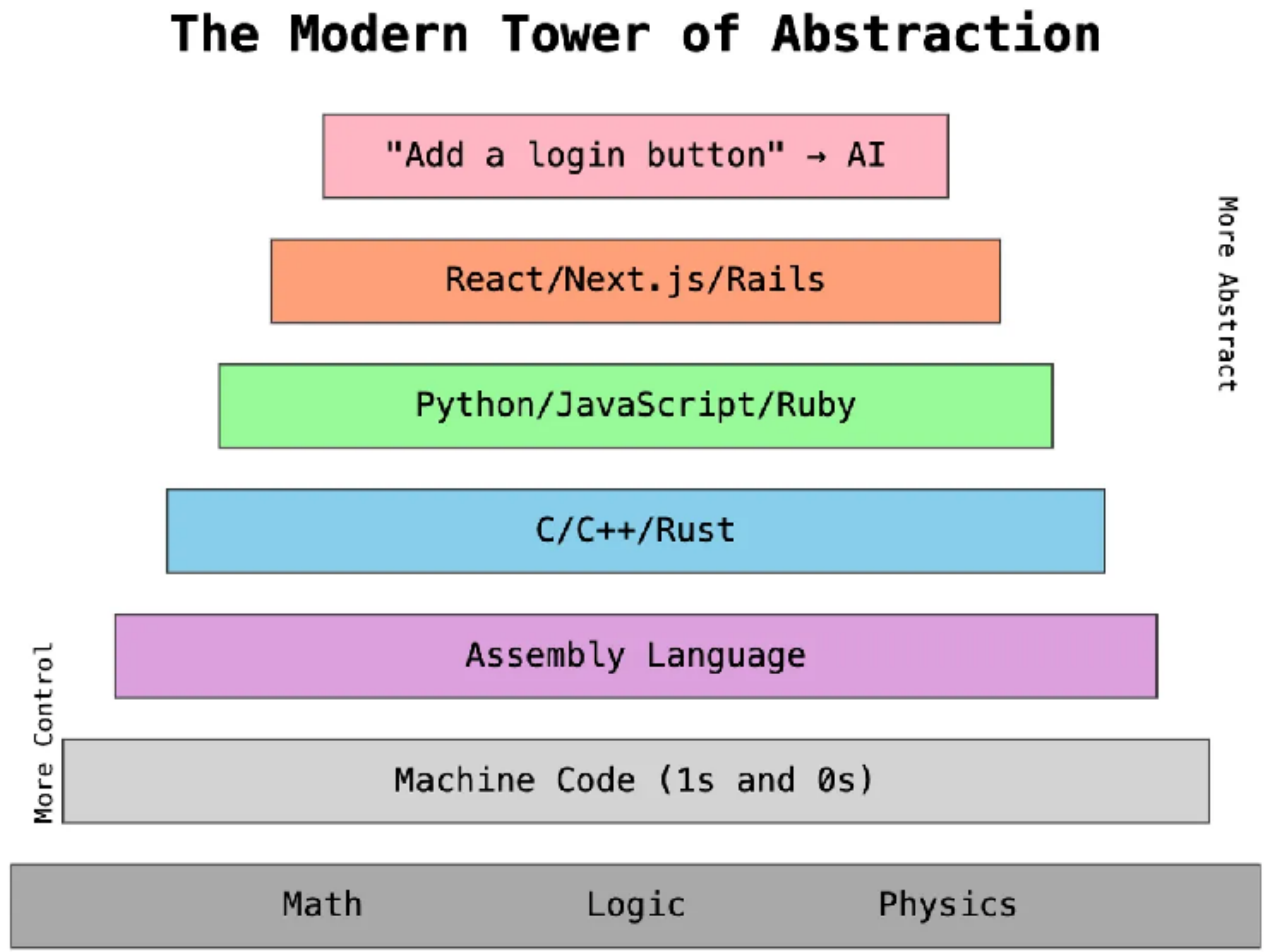

https://cline.bot/blog/from-assembly-to-ai-why-vibe-coding-is-just-another-chapter-in-our-abstraction-story

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Andrej Karpathy
@karpathy

There's a new kind of coding I call "vibe coding", where you fully give in to the vibes, embrace exponentials, and forget that the code even exists. It's possible because the LLMs (e.g. Cursor Composer w Sonnet) are getting too good. Also I just talk to Composer with SuperWhisper so I barely even touch the keyboard. I ask for the dumbest things like "decrease the padding on the sidebar by half" because I'm too lazy to find it. I "Accept All" always, I don't read the diffs anymore. When I get error messages I just copy paste them in with no comment, usually that fixes it. The code grows beyond my usual comprehension, I'd have to really read through it for a while. Sometimes the LLMs can't fix a bug so I just work around it or ask for random changes until it goes away. It's not too bad for throwaway weekend projects, but still quite amusing. I'm building a project or webapp, but it's not really coding - I just see stuff, say stuff, run stuff, and copy paste stuff, and it mostly works.

6:17 AM · Feb 3, 2025 · **5.3M** Views

https://x.com/karpathy/status/1886192184808149383

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Vibe Coding

Software development practice that uses artificial intelligence (AI) to generate functional code from **natural language prompts**, accelerating development, and making app building more accessible, especially for those with limited programming experience

https://cloud.google.com/discover/what-is-vibe-coding

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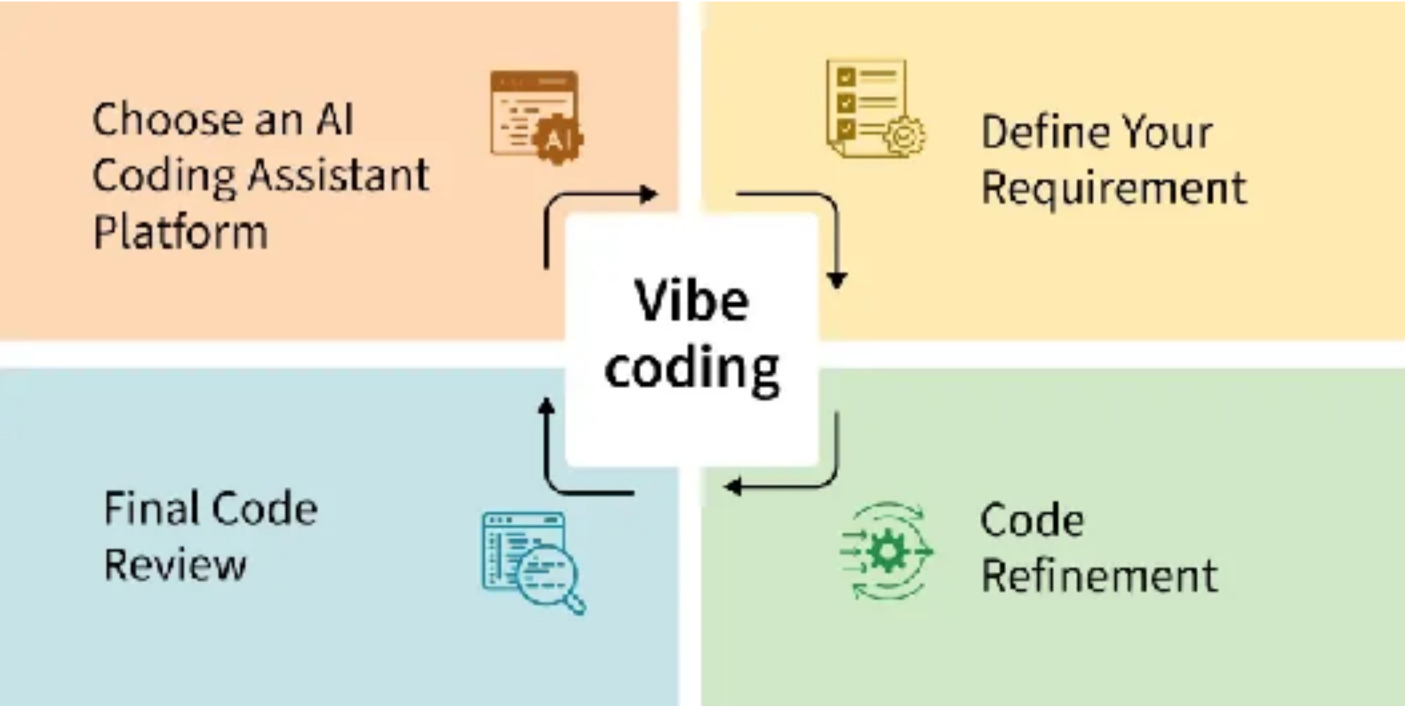
Types of Vibe Coding ?

Pure vibe coding

Responsible AI-assist development



https://cloud.google.com/discover/what-is-vibe-coding

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Working with AI Agent !!

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AI is the new normal in software development

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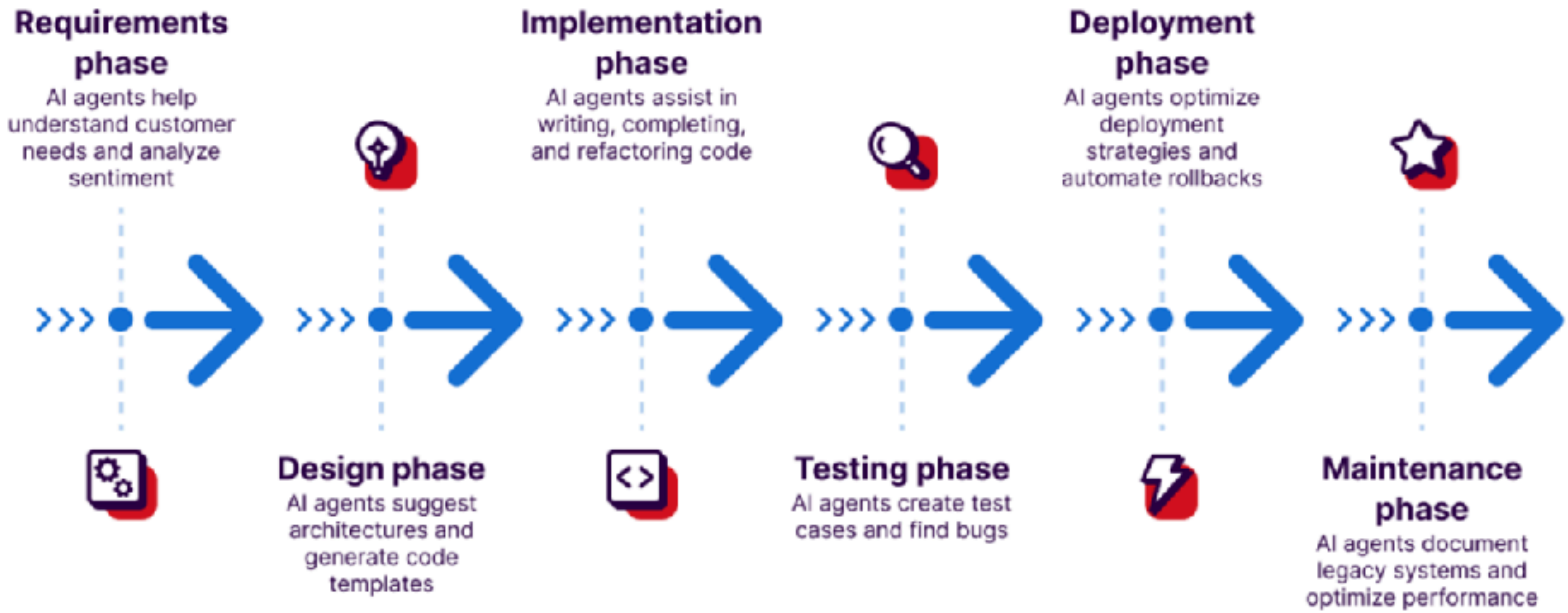
AI in Software Development



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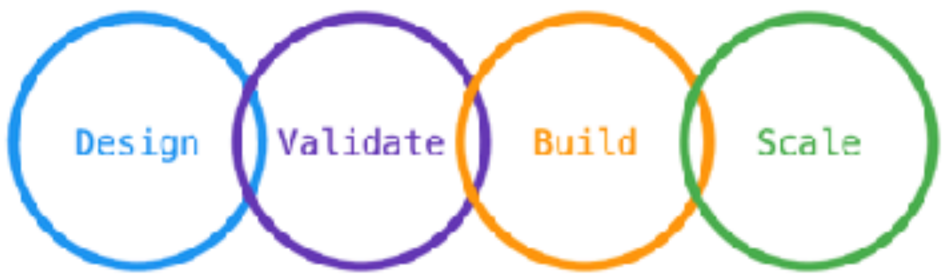
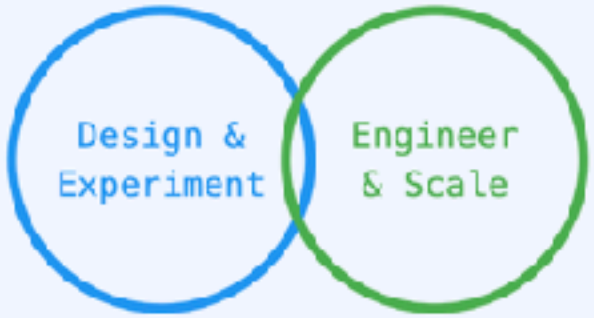
AI in Software Development

BEFORE



→ Fragmented SDLC Ownership;
→ Scaling Challenges;
→ Siloed Teams Increasing friction.

NOW



→ Democratized development;
→ Bridge between business and IT;
→ Composability for business agility.

AI-native SDLC drives faster, customer-centric, and higher-quality innovation.

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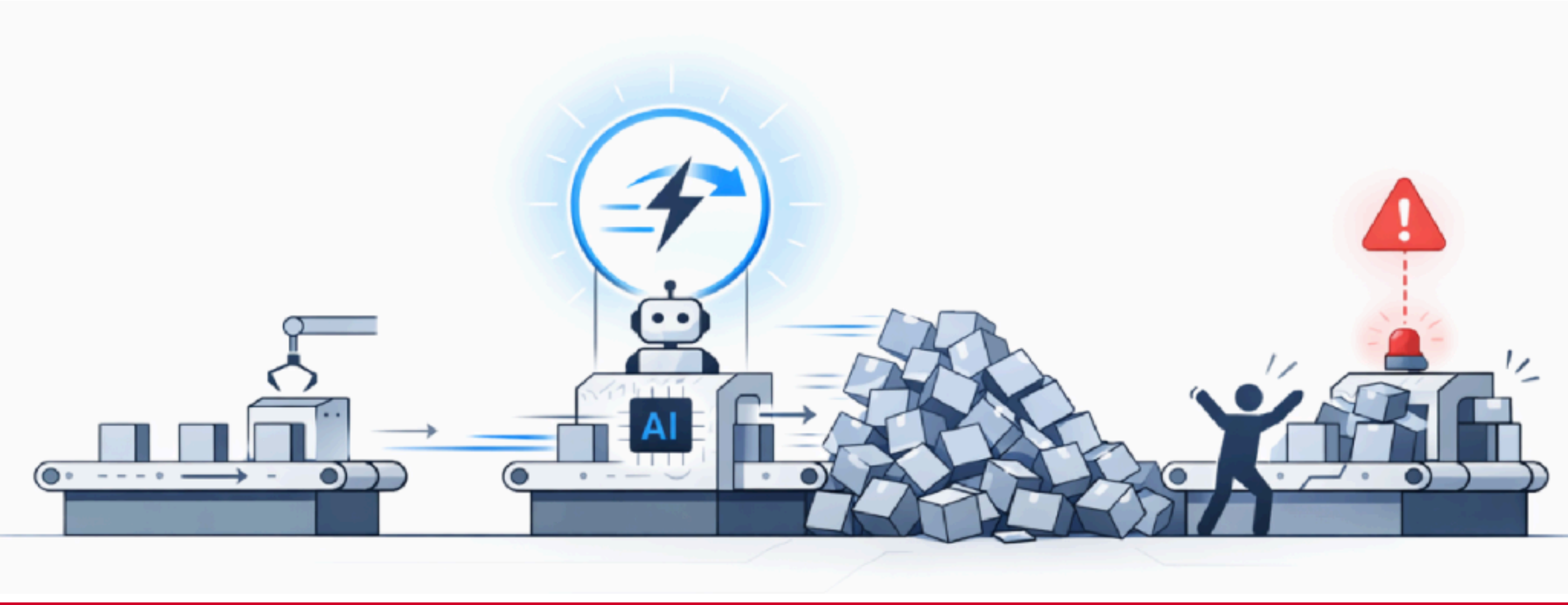
Invest in Real Bottleneck ?



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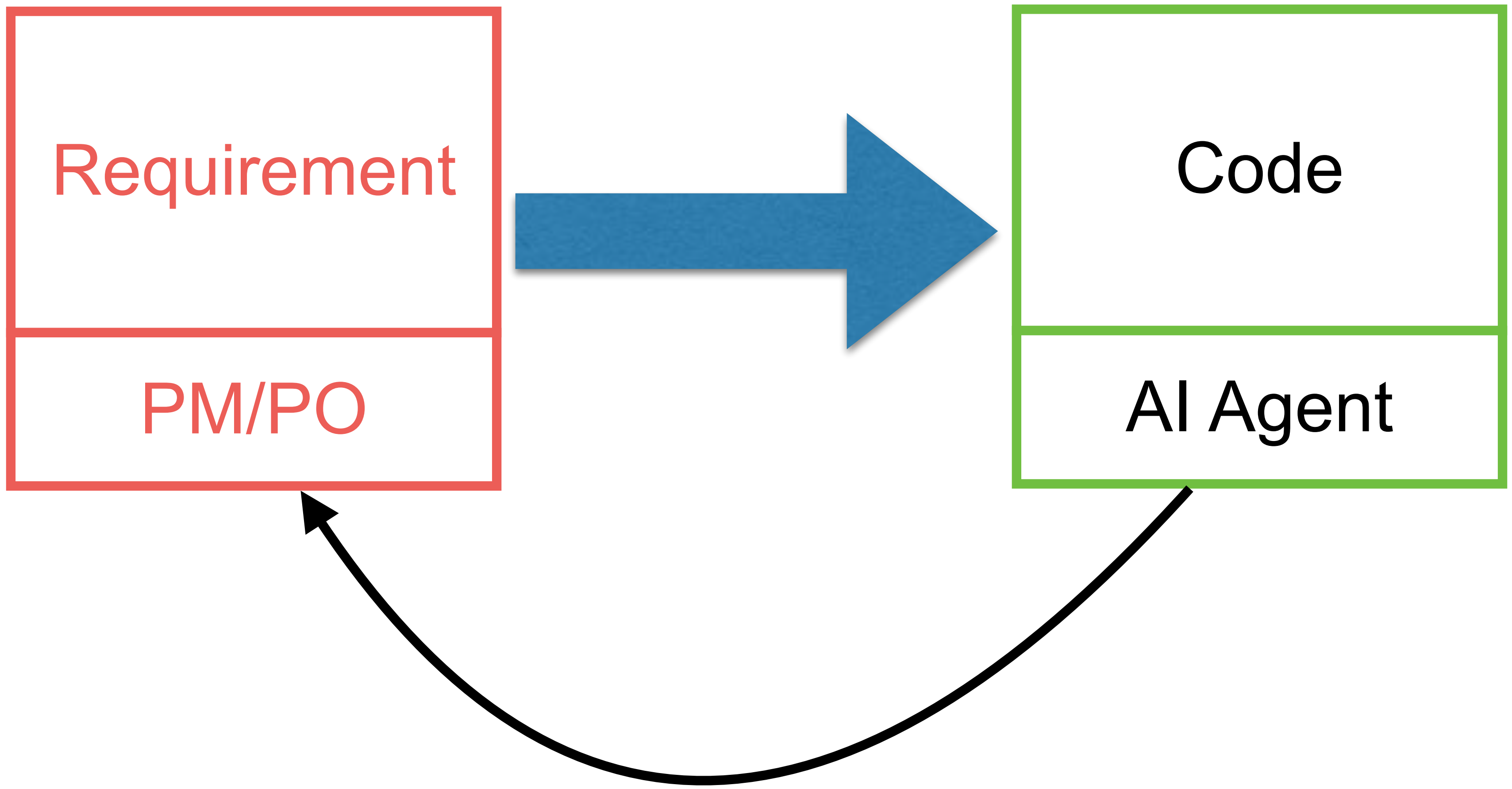
New Bottleneck ?

Decide what to **build**, not code



Get feedback faster

https://www.theaiopportunities.com/p/is-ai-slowing-down-the-new-bottleneck

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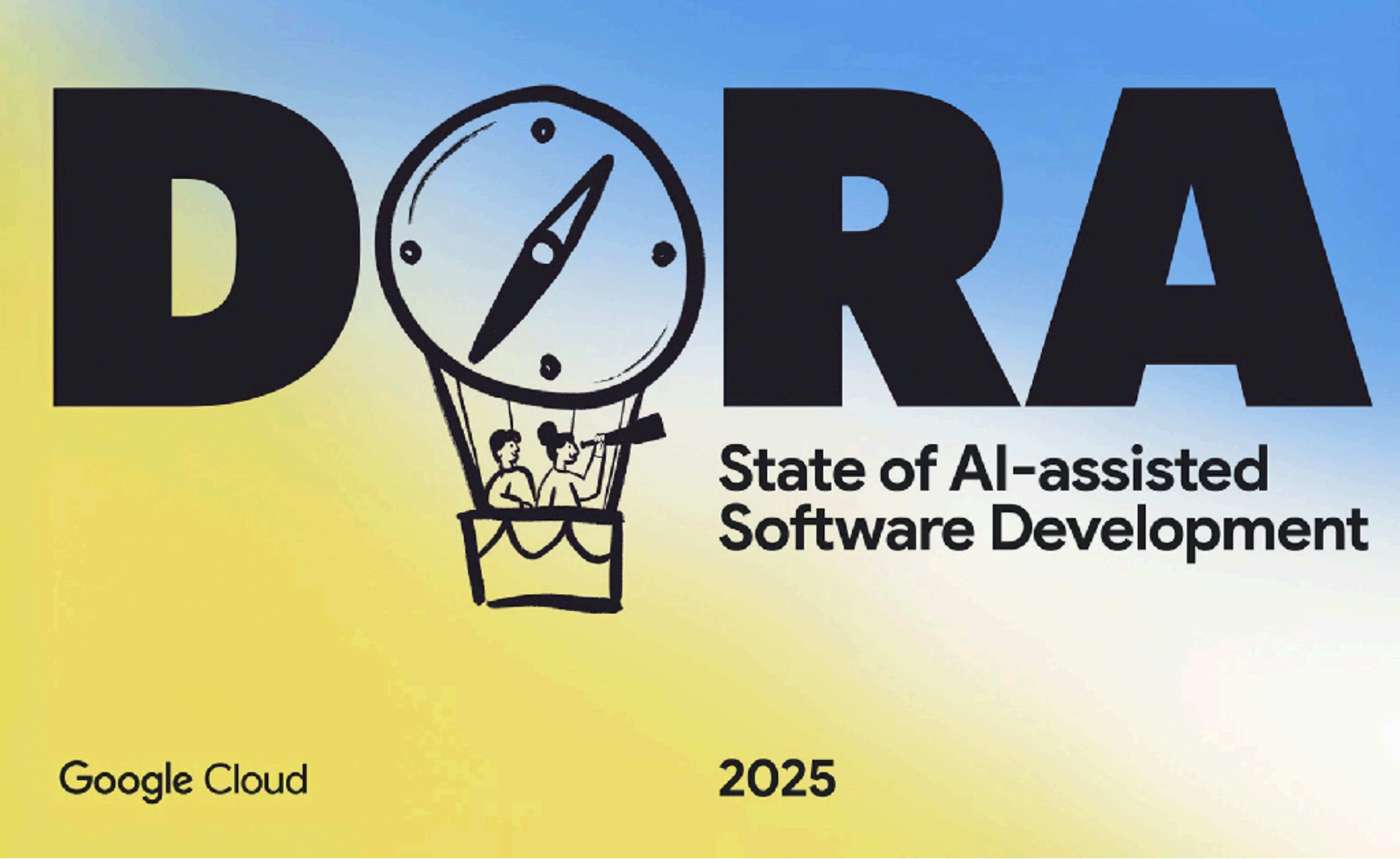

https://cloud.google.com/blog/products/ai-machine-learning/announcing-the-2025-dora-report

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AI and Productivity ?

AI Adoption

90% of software development professionals are now using AI at work

Productivity

80% report that AI has increased their productivity

Reliance

65% are relying on AI at least a “moderate amount” at work

Code Quality

59% of survey respondents say that AI has positively impacted their code quality

https://blog.google/innovation-and-ai/technology/developers-tools/dora-report-2025/

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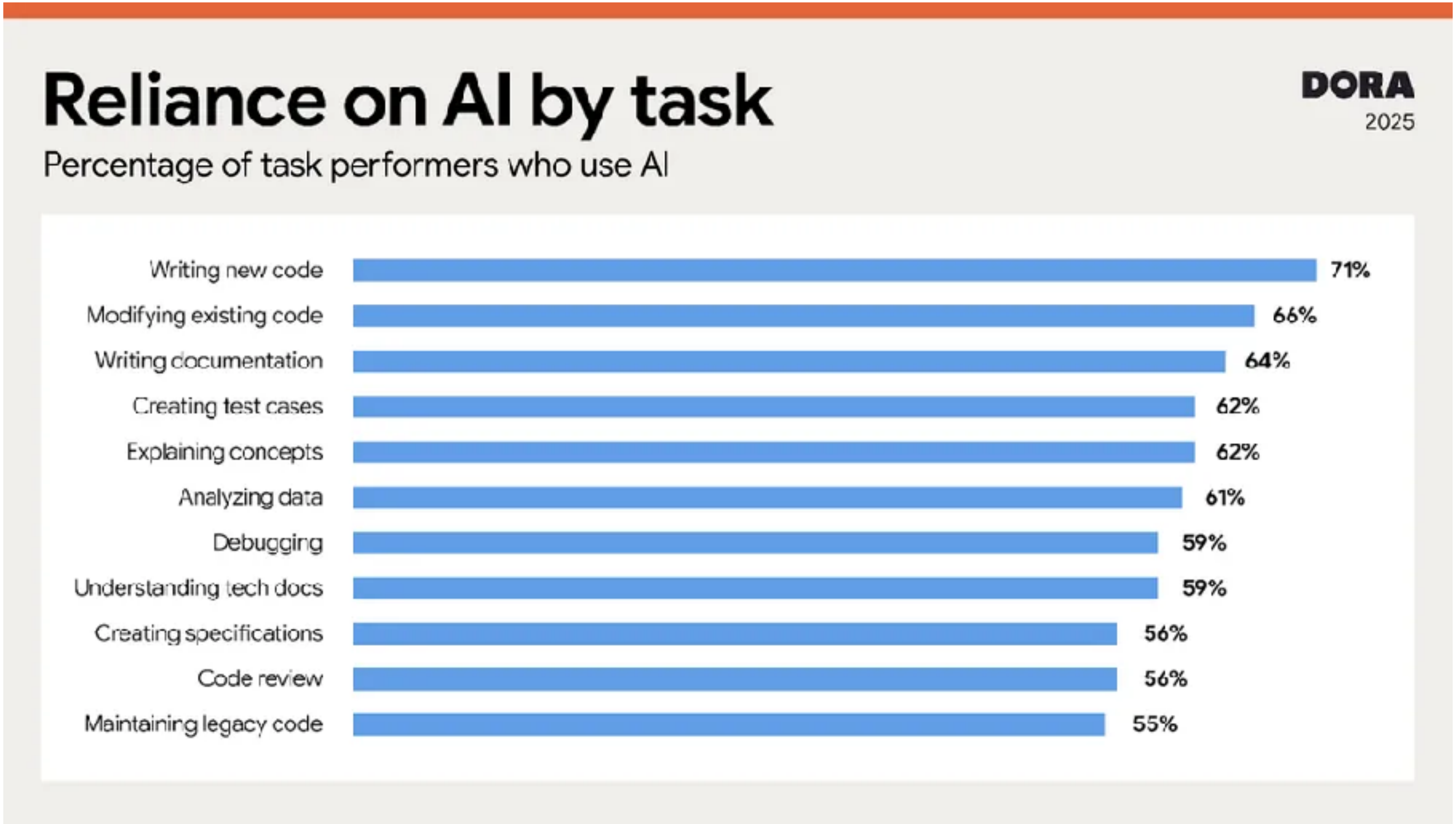
Productivity ?



https://blog.google/innovation-and-ai/technology/developers-tools/dora-report-2025/

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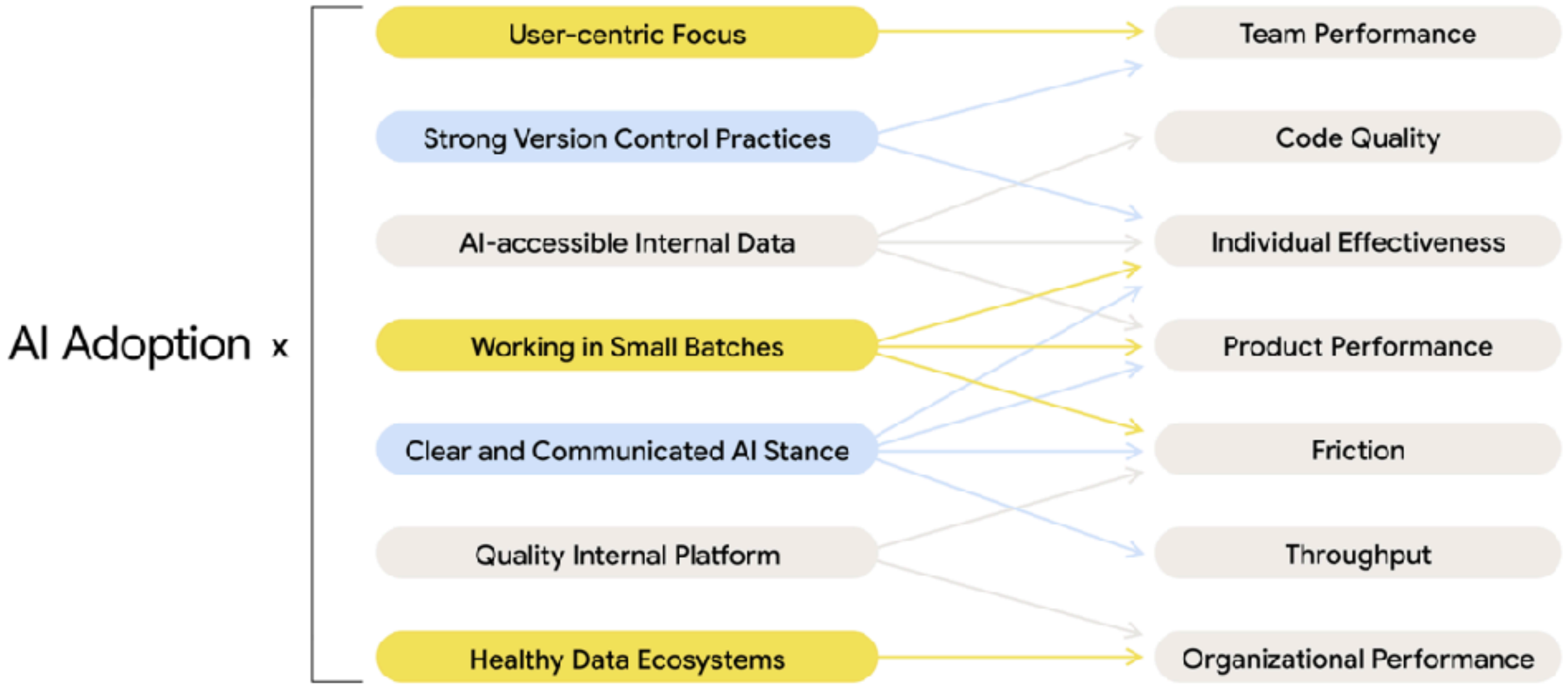
DORA AI Capabilities Model



https://cloud.google.com/blog/products/ai-machine-learning/announcing-the-2025-dora-report

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AI Agent for Coding

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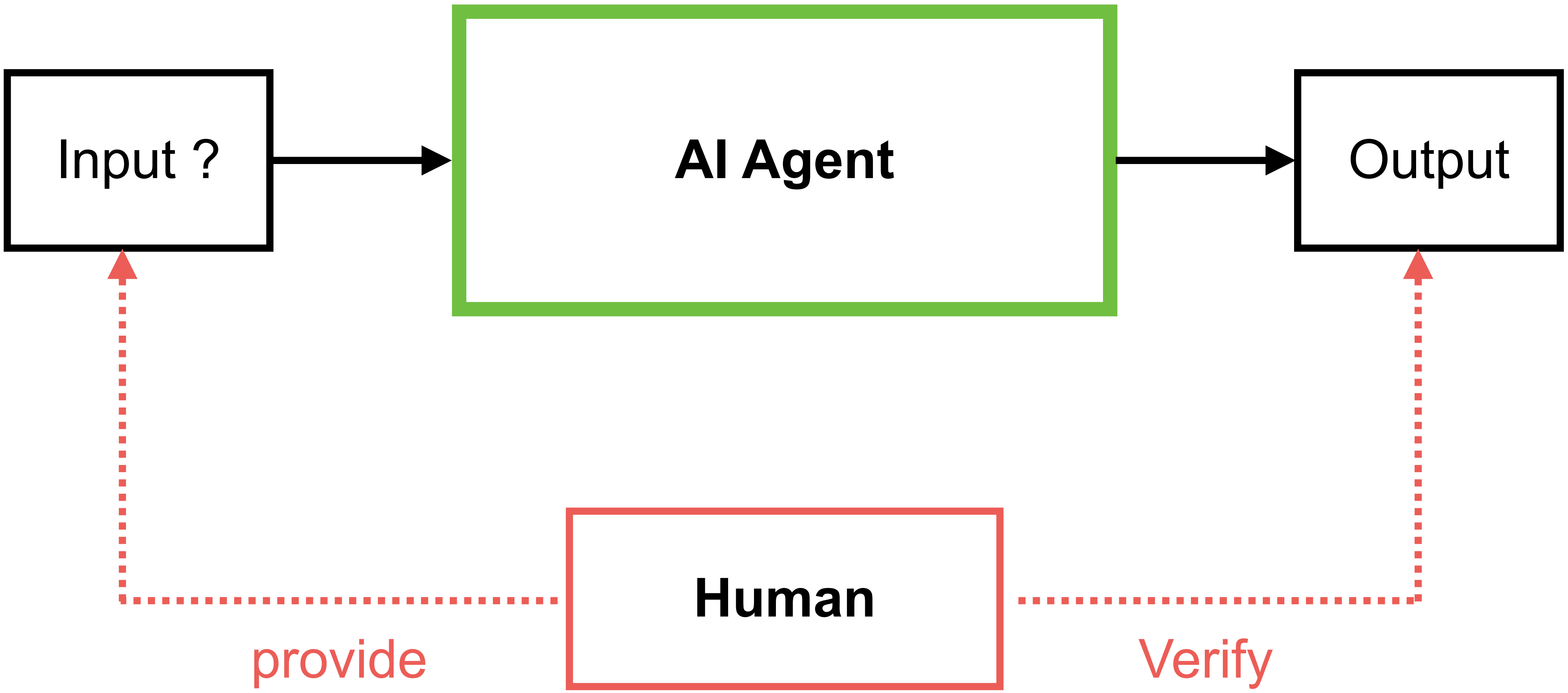
AI Agent + Human in/on the Loop



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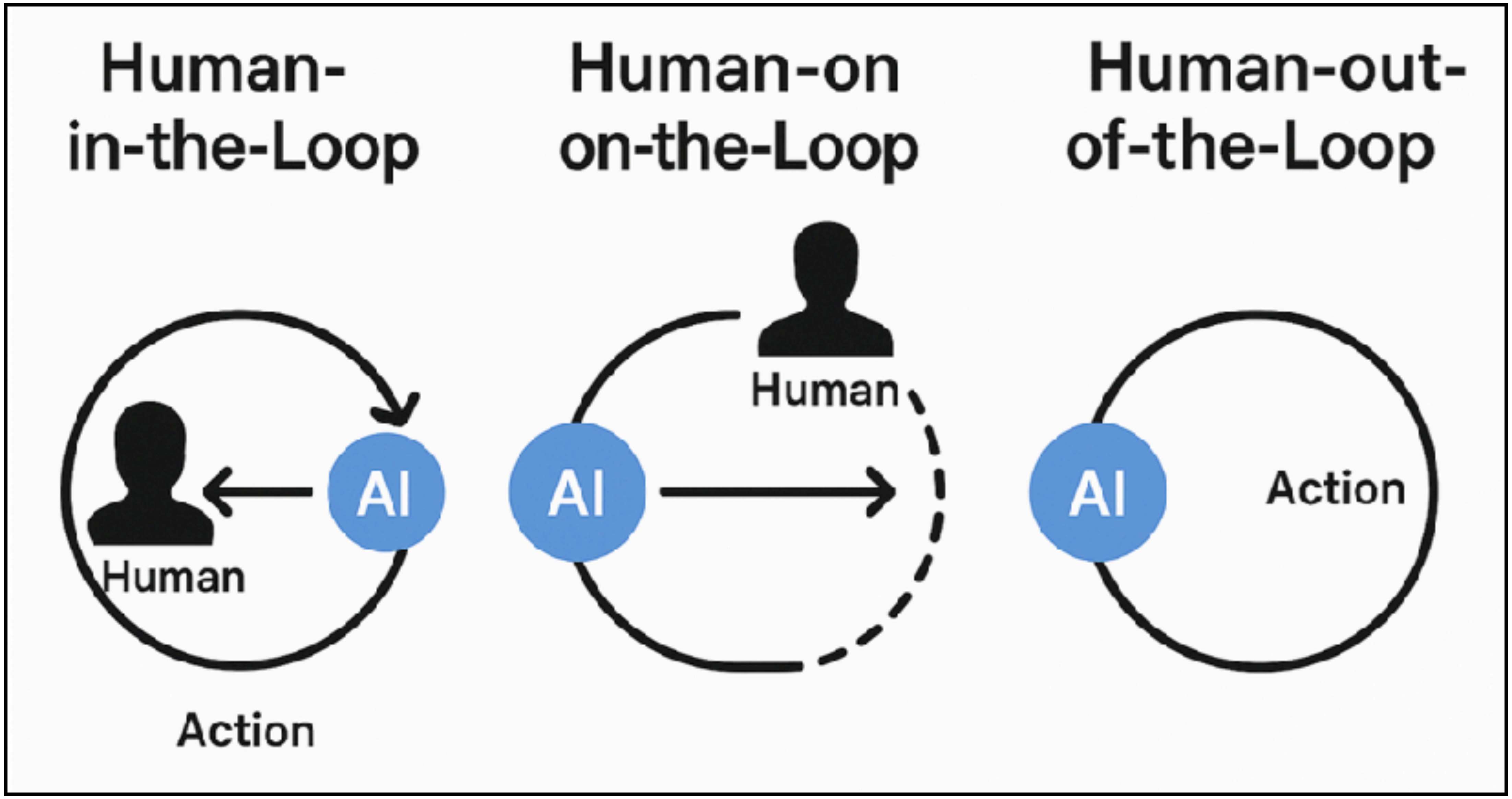
AI Agent with/without Human



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AI draft
Human craft

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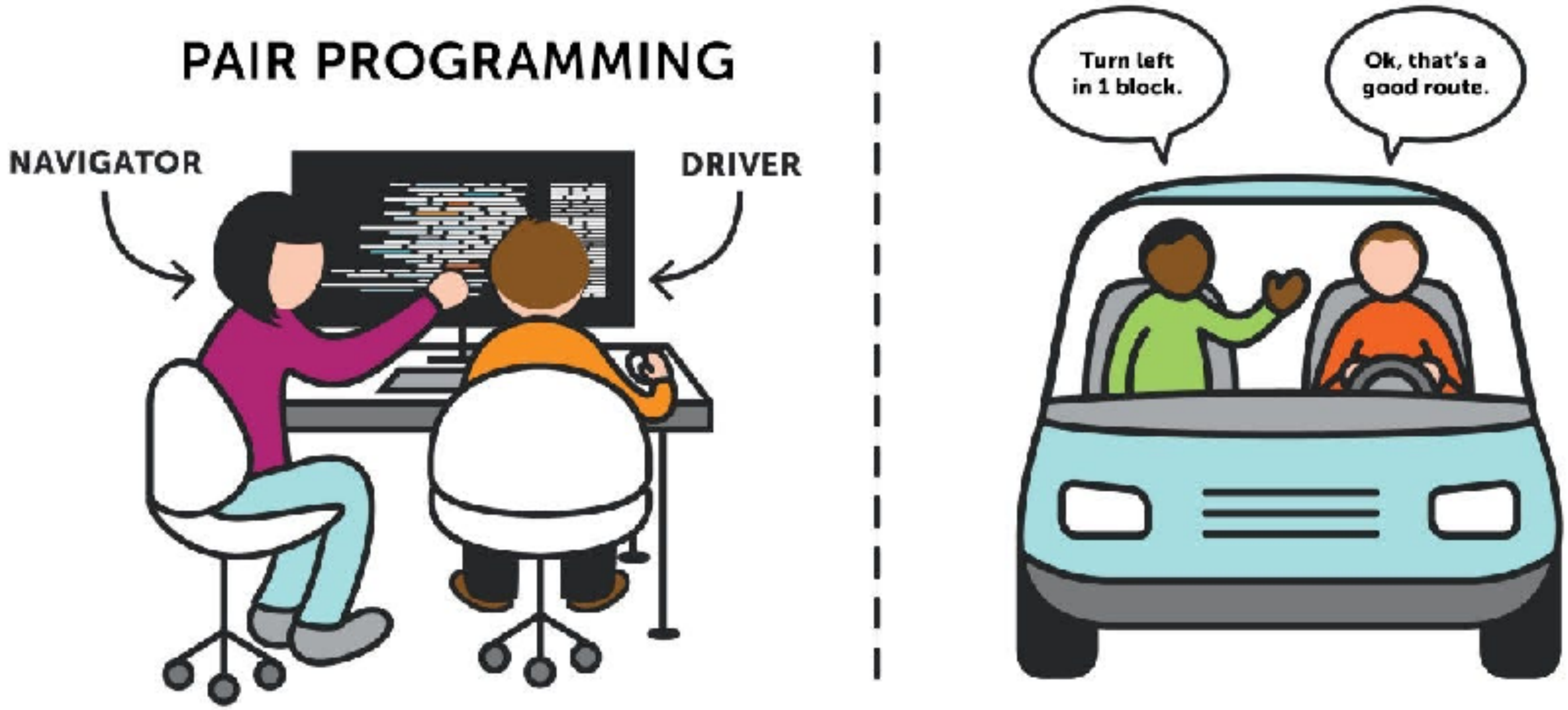

Reviews, **Tests** and **Understands**
with experience

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AI Agent for Coding

AI Agent

Controller loop (reasoning loop)

LLM

Code writing tools

Memory and Context

Other tools

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How to work with AI Agent ?

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Prompt Engineering
(what to ask ?)

Context Engineering
(what to send to model ?)

Harness Engineering
(how the whole thing operate ?)

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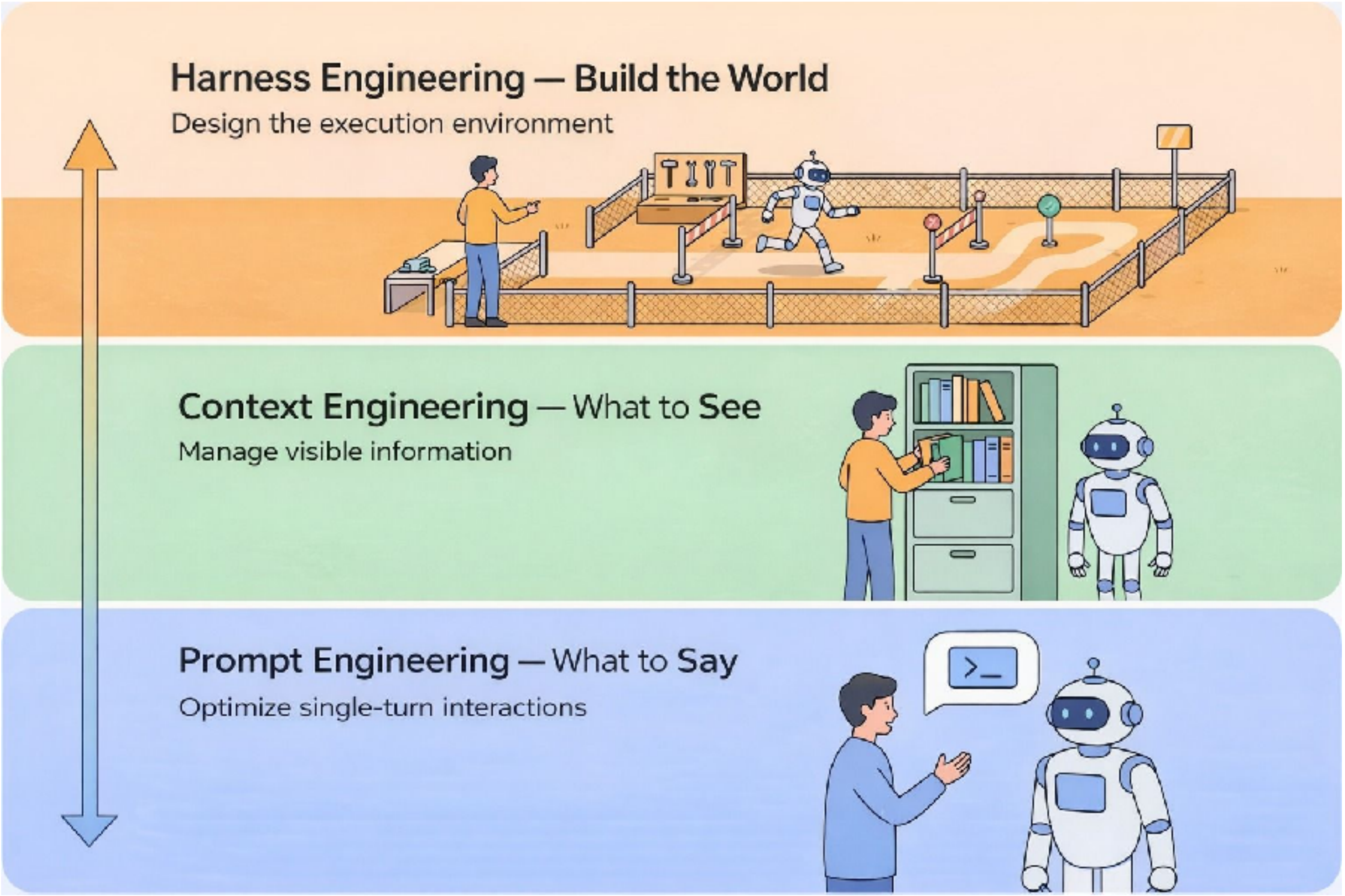

https://milvus.io/blog/harness-engineering-ai-agents.md

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2 Background: From Weights to Context to Harness

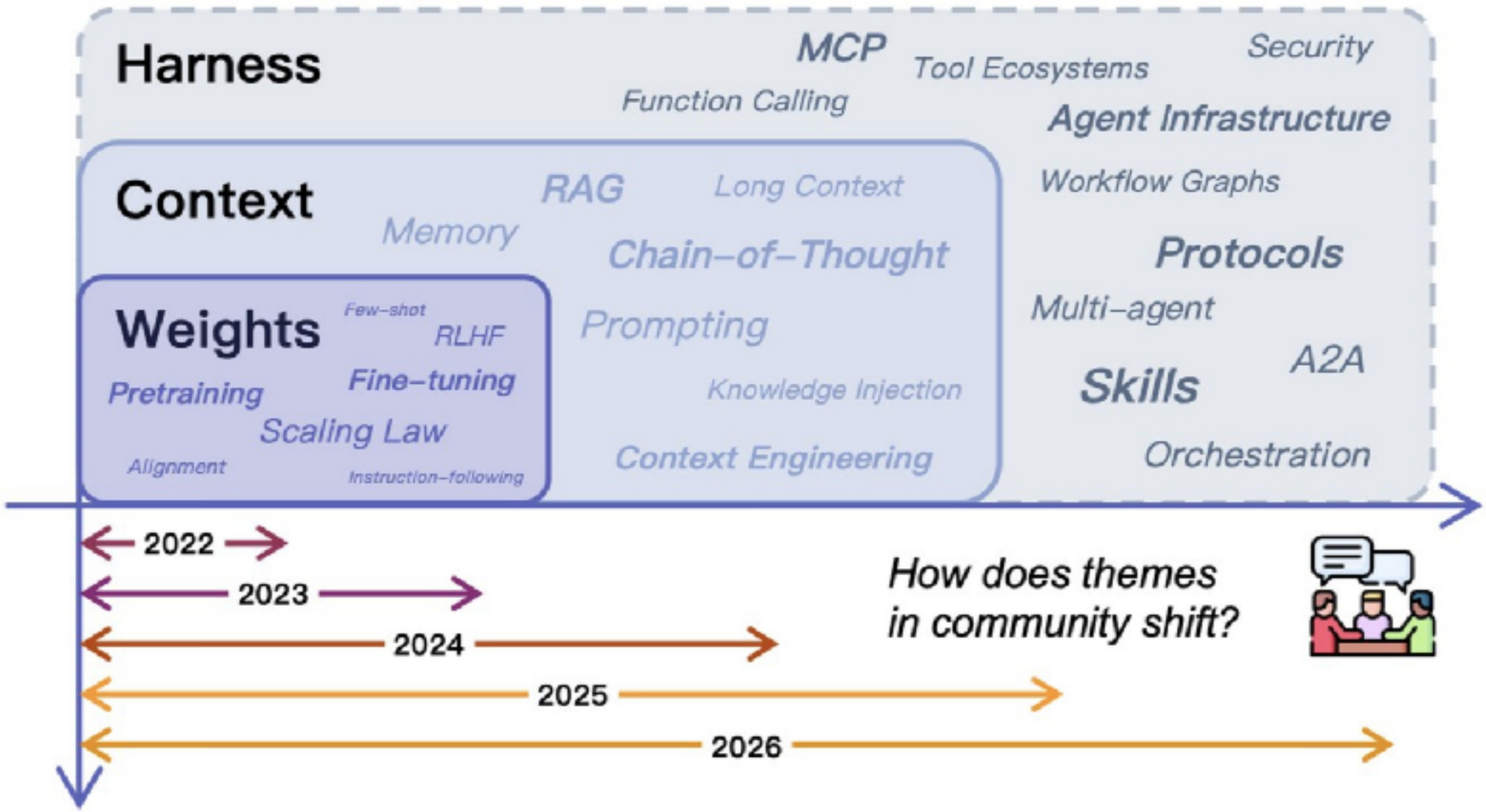

Figure 2 Community theme evolution across three capability layers. The stacked layers—Weights, Context, and Harness—show how the center of gravity in the LLM agent community has shifted outward over time, from parametric knowledge and prompting toward harness-level infrastructure such as tool ecosystems, protocols, skills, and multi-agent orchestration.

https://arxiv.org/pdf/2604.08224

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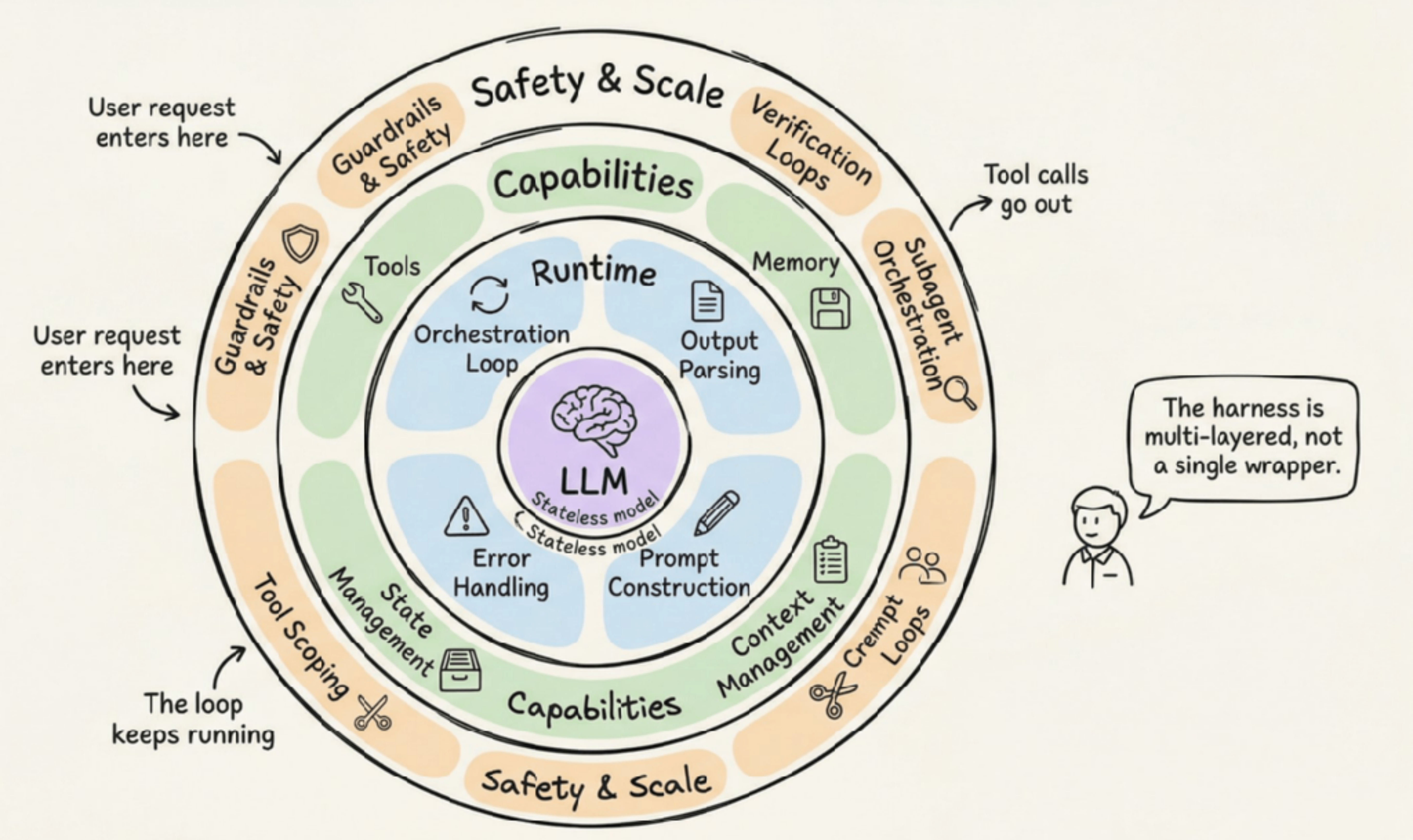

https://blog.dailydoseofds.com/p/evolution-of-agent-landscape-from

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AI Agent = LLM + Harness

Harness = AI Agent - LLM

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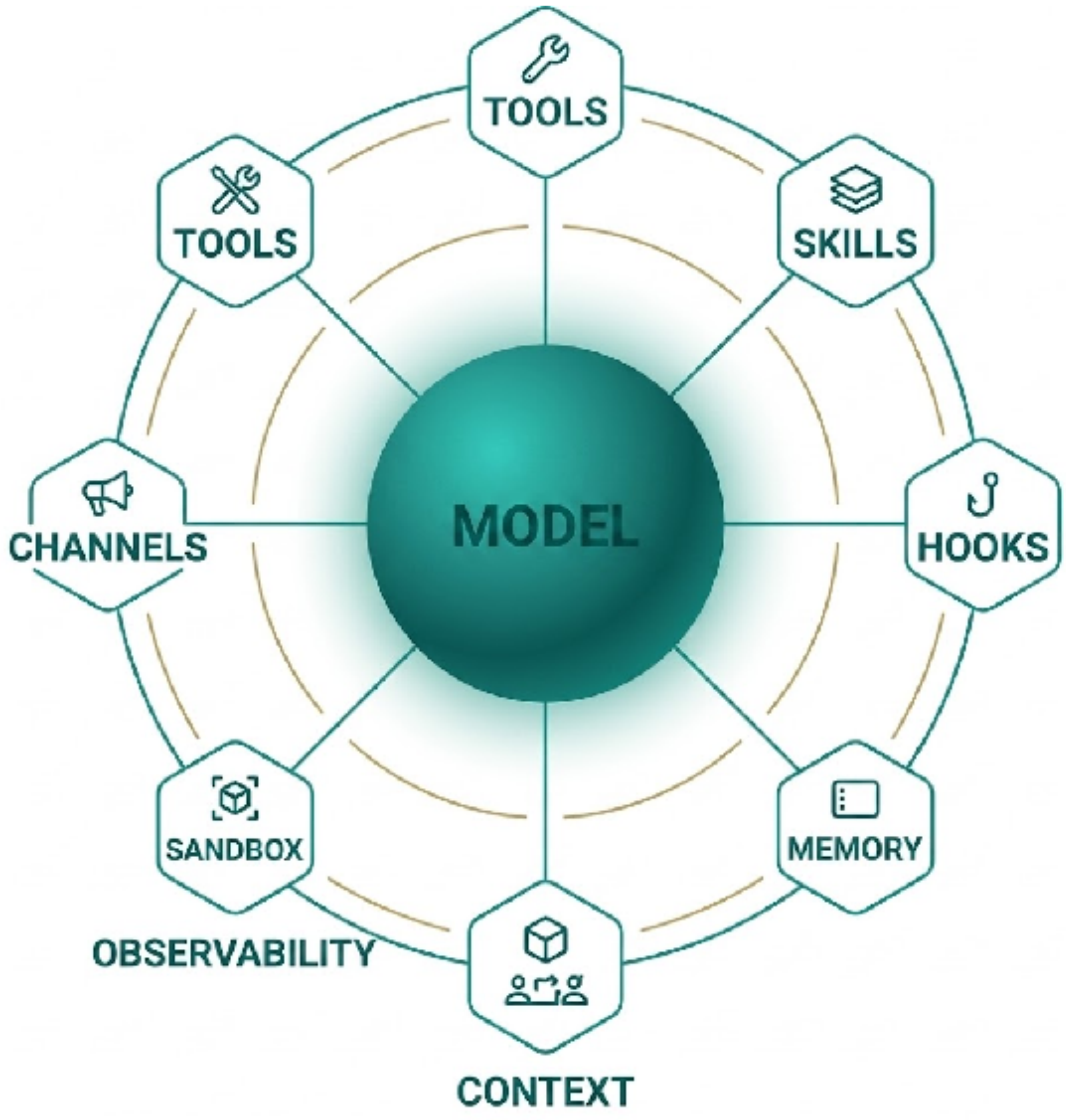

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Workflows with AI Agent ?

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Workflows ?

Application level
Code level

https://cloud.google.com/discover/what-is-vibe-coding

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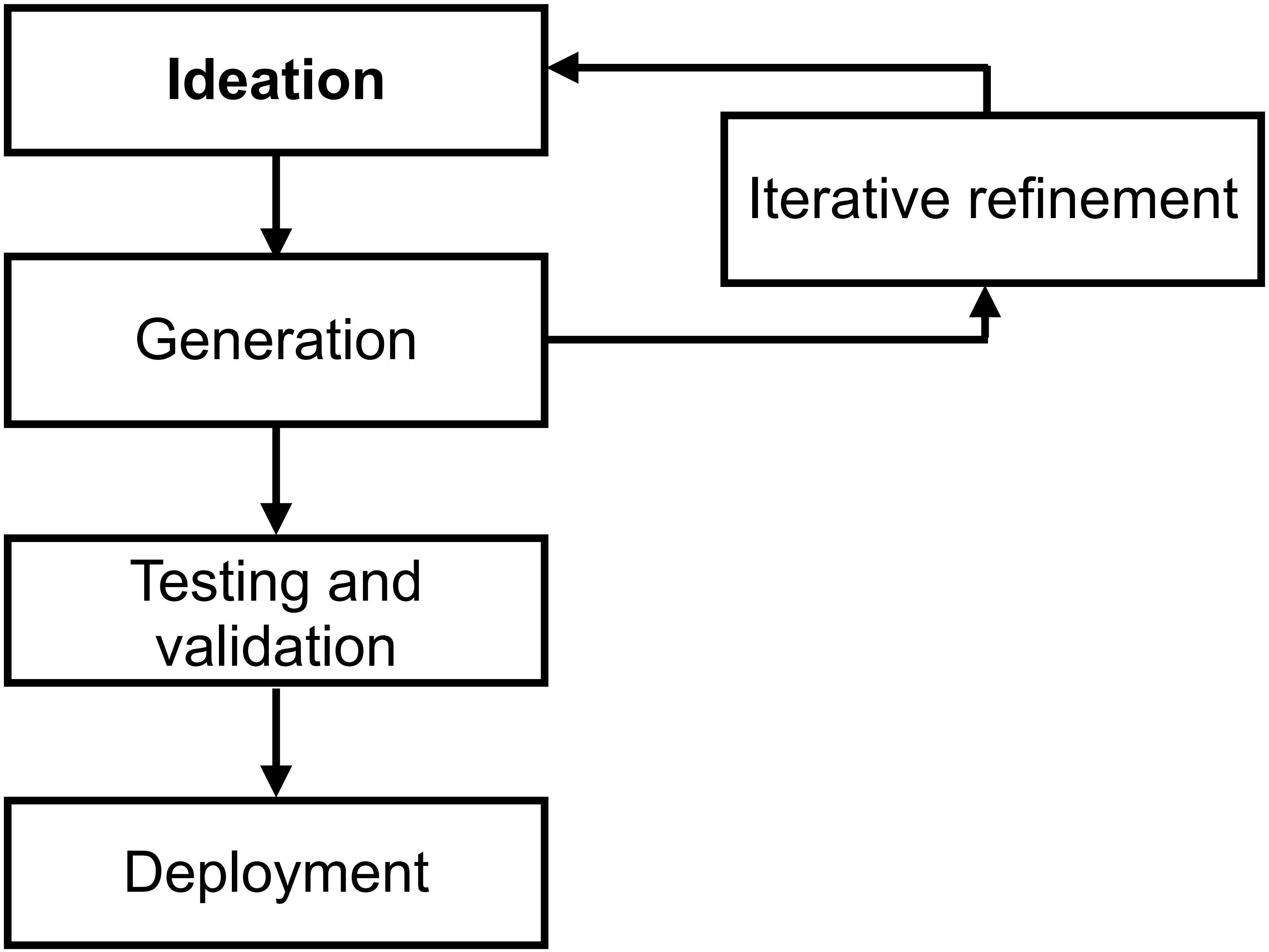
Application level workflow

Talking about **high-level idea** from concept to deployment



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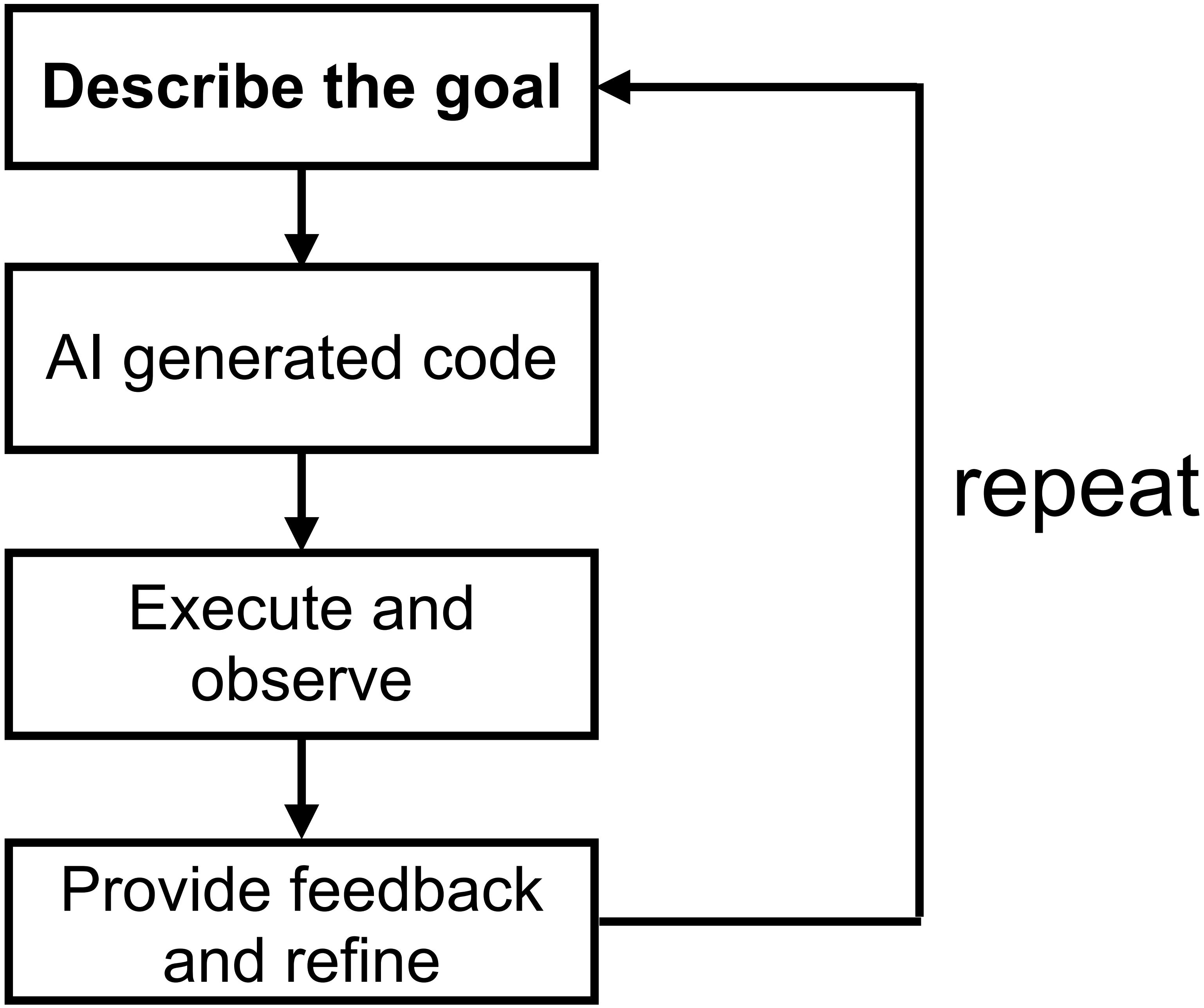
Code level workflow

Conversational loop to create a **specific code**



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Tools ?

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AI Agent for Coding !!

CLAUDE CODE

GitHub Copilot
Visual Studio Code

Antigravity

cline

windsurf

opencode

CURSOR

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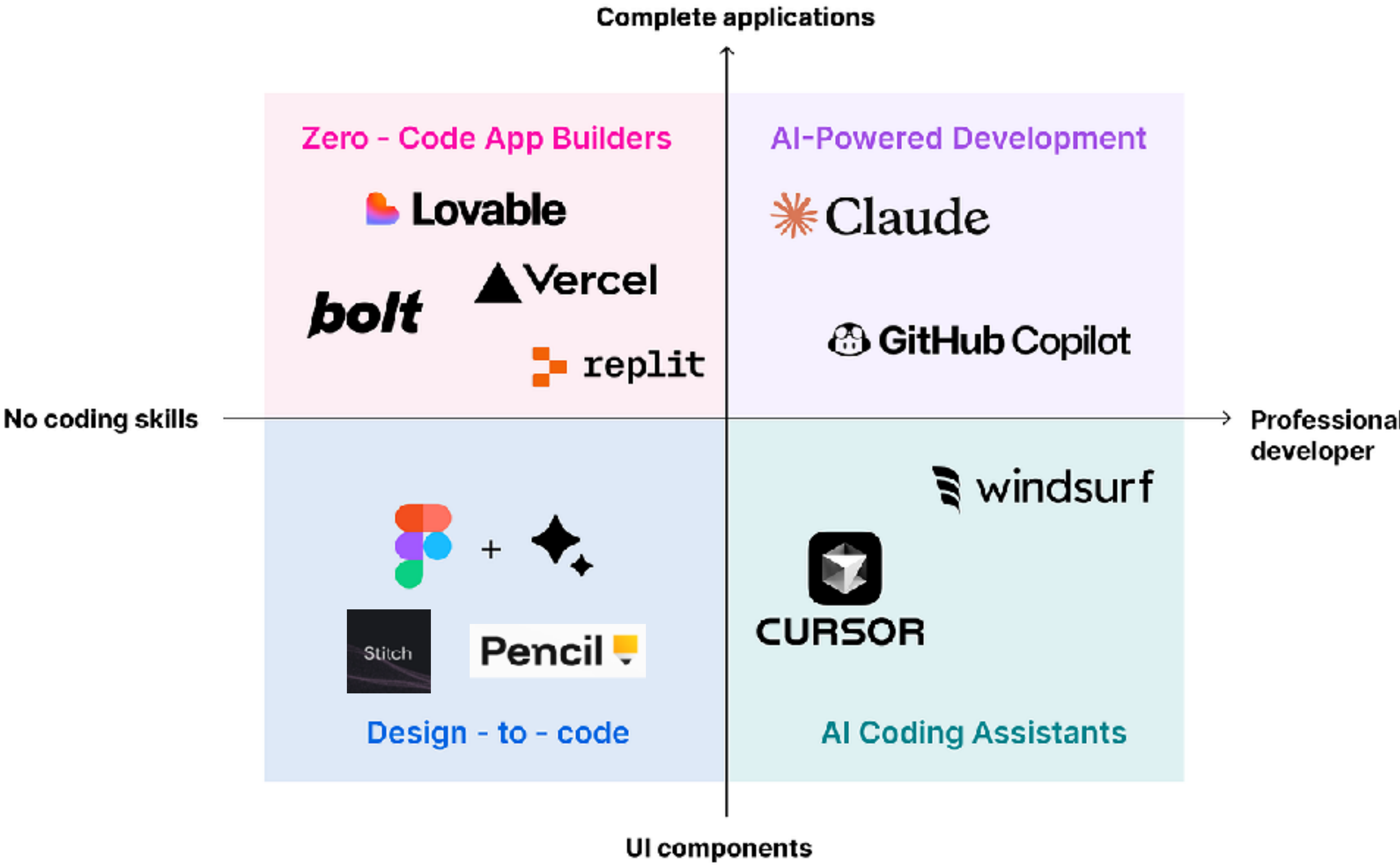

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Before start …

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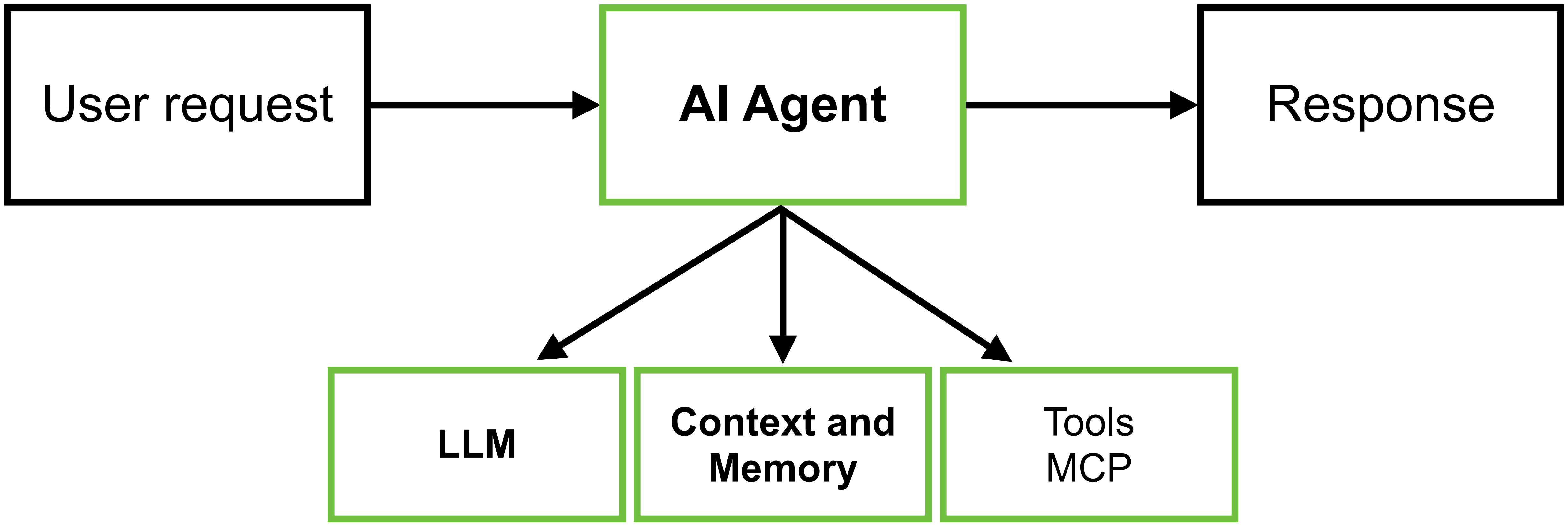
AI Agent for Coding



How to manage context or knowledge for Agent ?

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Token is new oil !!
Token is expensive !!

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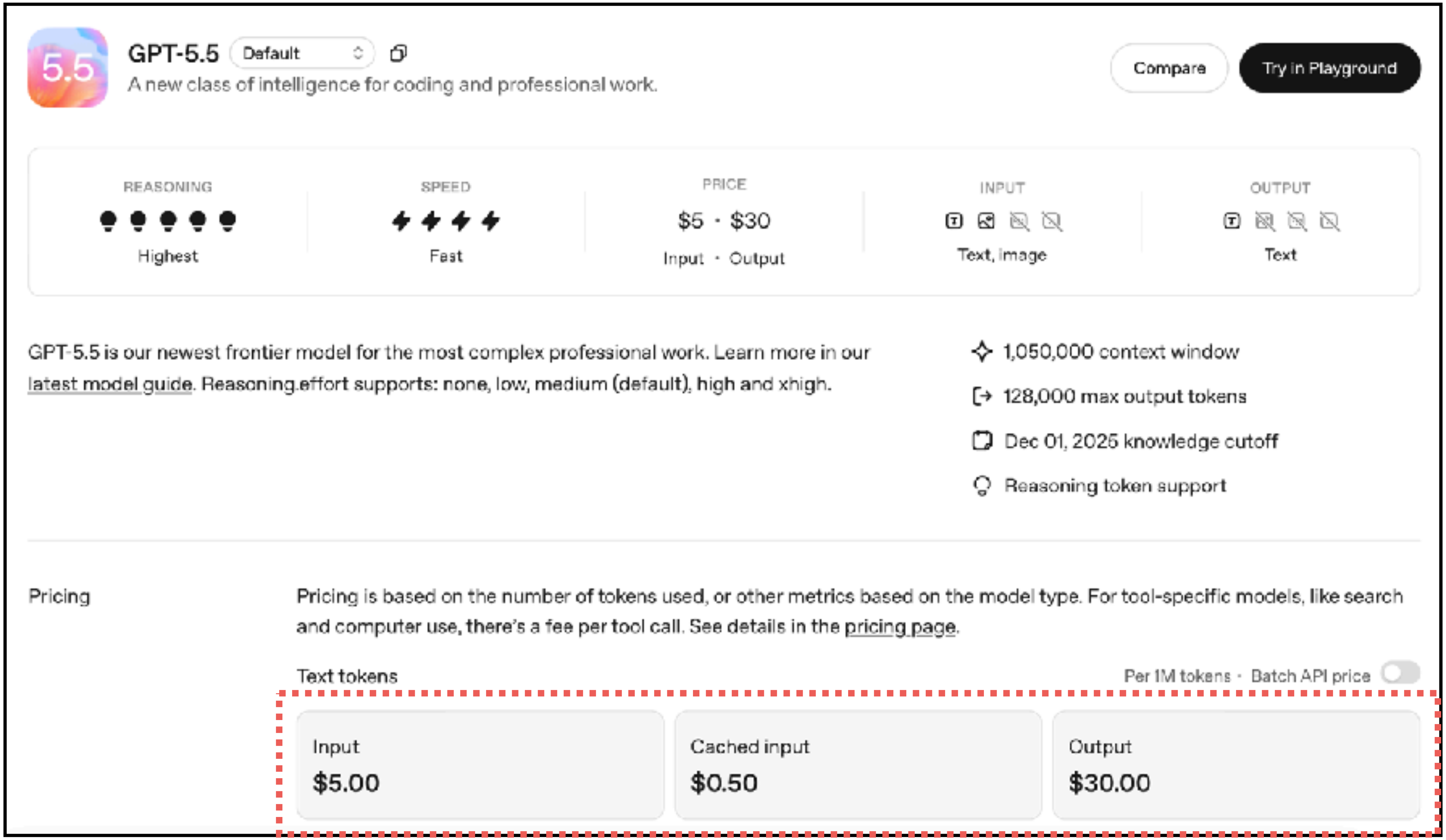
LLM Model :: GPT 5.5



https://developers.openai.com/api/docs/models/gpt-5.5

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LLM Model :: Anthropic

Model	Base Input Tokens	5m Cache Writes	1h Cache Writes	Cache Hits & Refreshes	Output Tokens
Claude Opus 4.7	$5 / MTok	$6.25 / MTok	$10 / MTok	$0.50 / MTok	$25 / MTok
Claude Opus 4.6	$5 / MTok	$6.25 / MTok	$10 / MTok	$0.50 / MTok	$25 / MTok
Claude Opus 4.5	$5 / MTok	$6.25 / MTok	$10 / MTok	$0.50 / MTok	$25 / MTok
Claude Opus 4.1	$15 / MTok	$18.75 / MTok	$30 / MTok	$1.50 / MTok	$75 / MTok
Claude Opus 4	$15 / MTok	$18.75 / MTok	$30 / MTok	$1.50 / MTok	$75 / MTok
Claude Sonnet 4.6	$3 / MTok	$3.75 / MTok	$6 / MTok	$0.30 / MTok	$15 / MTok
Claude Sonnet 4.5	$3 / MTok	$3.75 / MTok	$6 / MTok	$0.30 / MTok	$15 / MTok
Claude Sonnet 4	$3 / MTok	$3.75 / MTok	$6 / MTok	$0.30 / MTok	$15 / MTok

https://platform.claude.com/docs/en/about-claude/pricing

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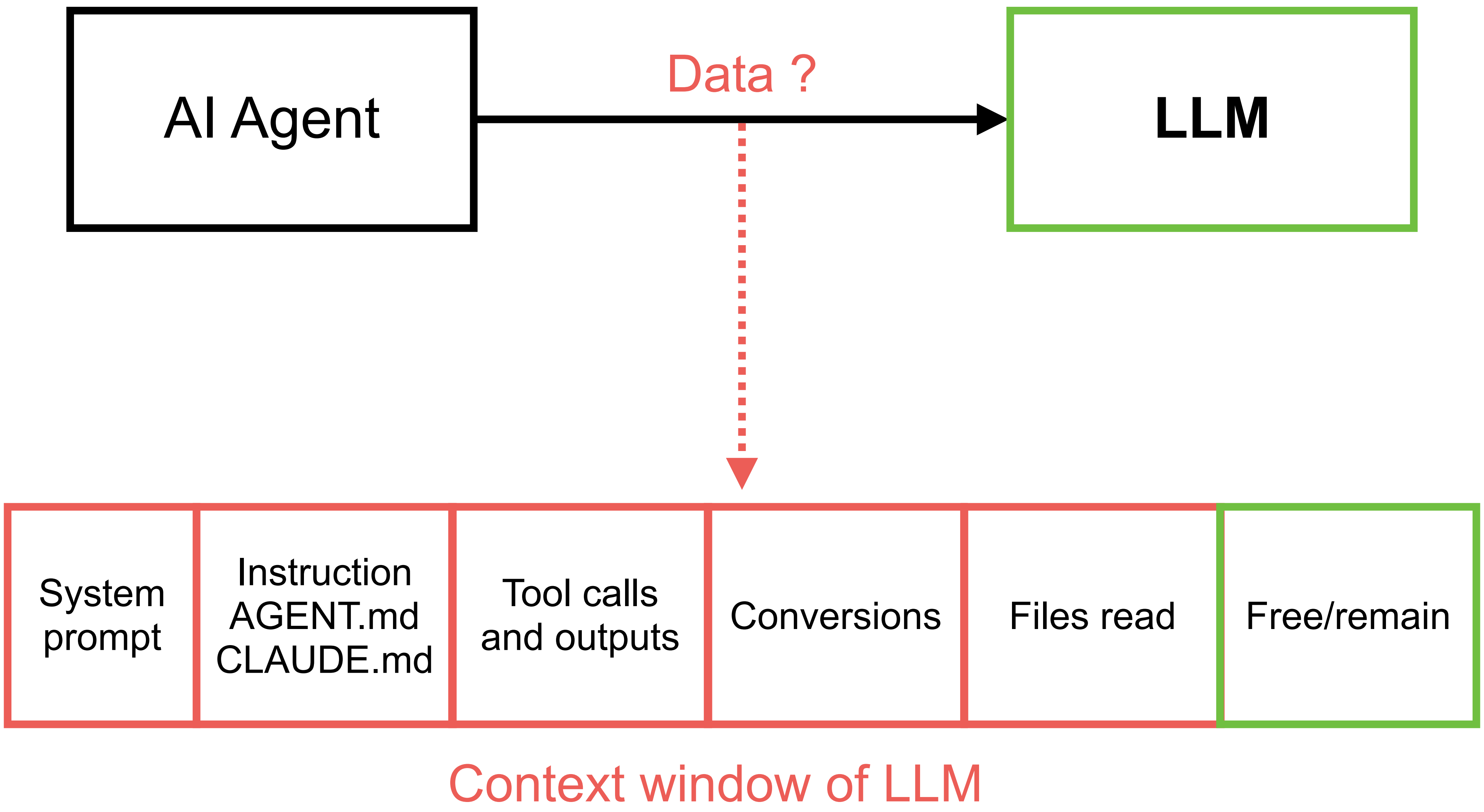
Context Window ?



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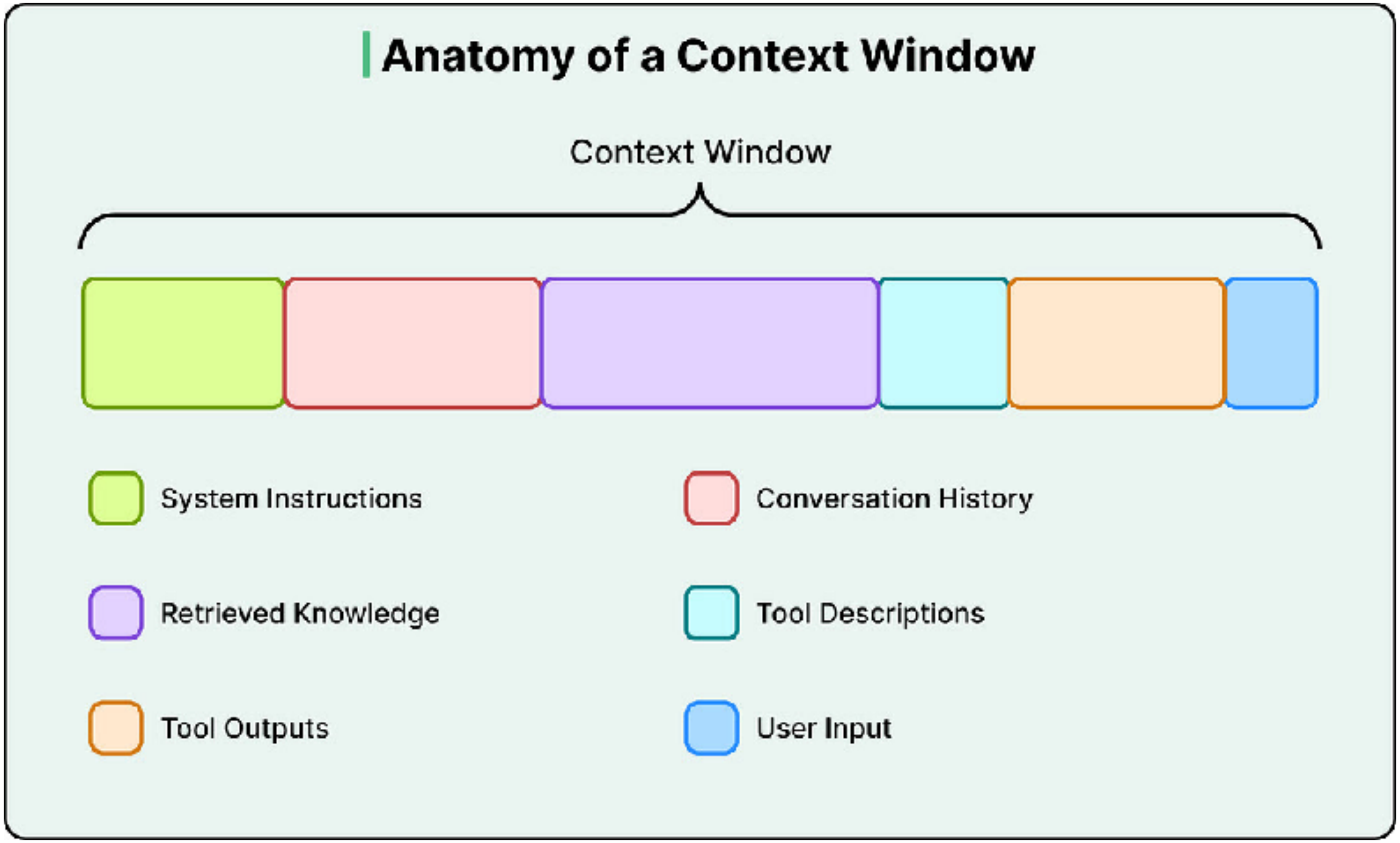
Anatomy of Context Window



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Types of Knowledge/Context ?

Shared context

Static Knowledge

Prompt engineering

Global Instructions and rules

AGENTS.md

CLAUDE.md

DESIGN.md

Task context

Dynamic Knowledge

Memory

Graph/Vector database

MCP tools

Specific Instructions

Agent SKILLs

Sub-agent and Modes

Context engineering ?

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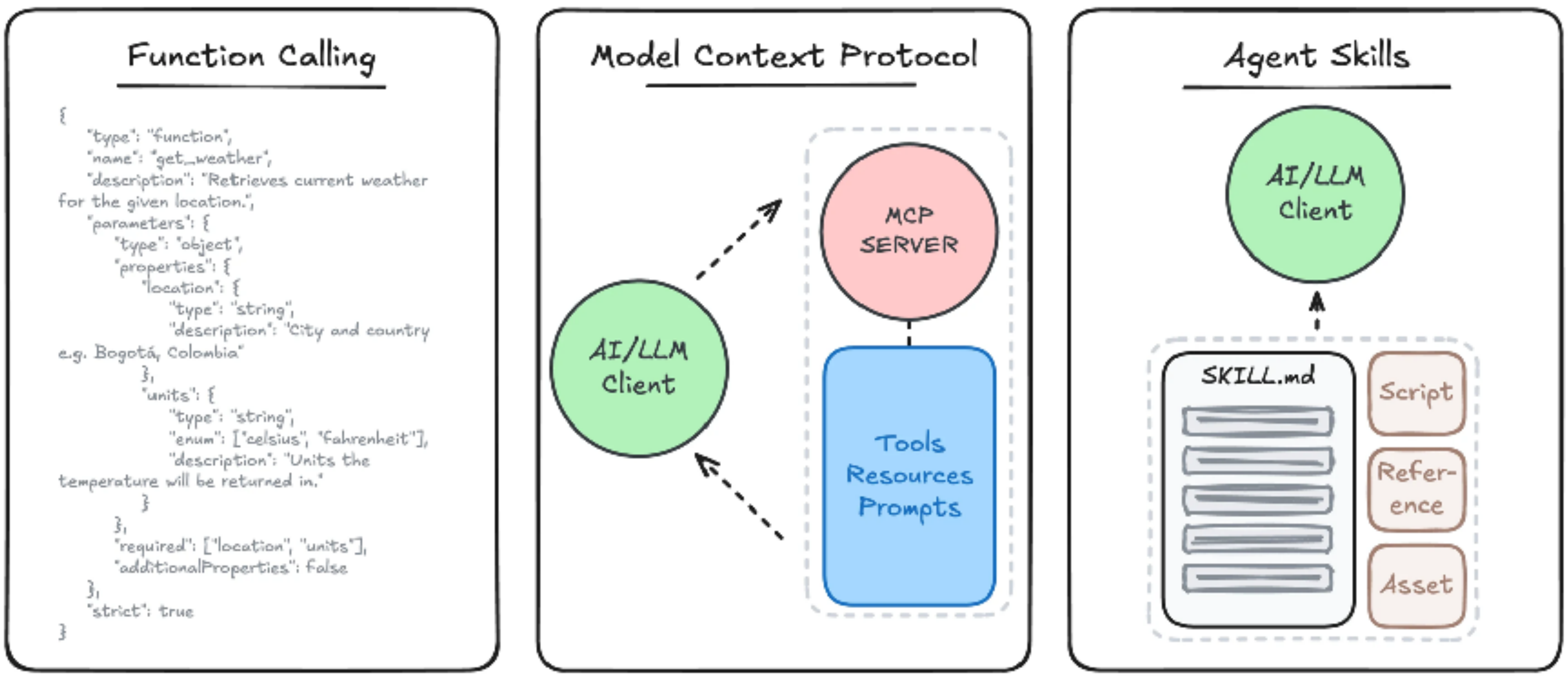
Agent Skills

Standard way to give AI agent new capabilities and expertise



https://agentskills.io/home

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How to manage context ?

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Context Engineering

Right information

Right tools

Structured context

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Long context == Better result ?

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Context window size of models

Different models may have different window size

Model name	Context window size
GPT 5.5	1M
Gemini 1.5, 2.5	1M
Llama 4	1M
Claude Opus 4.7	1M

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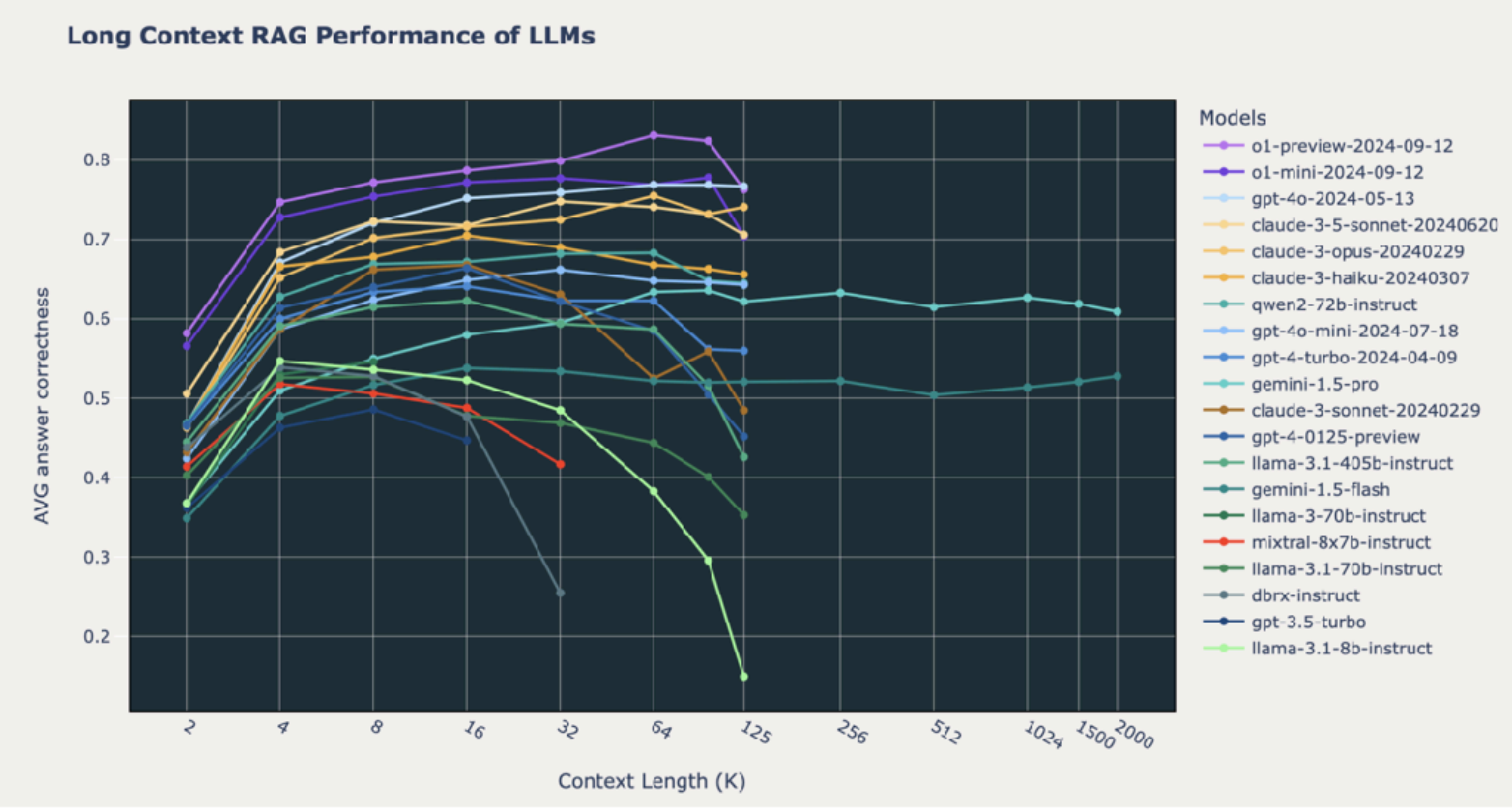
Long context LLM is better !!



https://arxiv.org/abs/2411.03538

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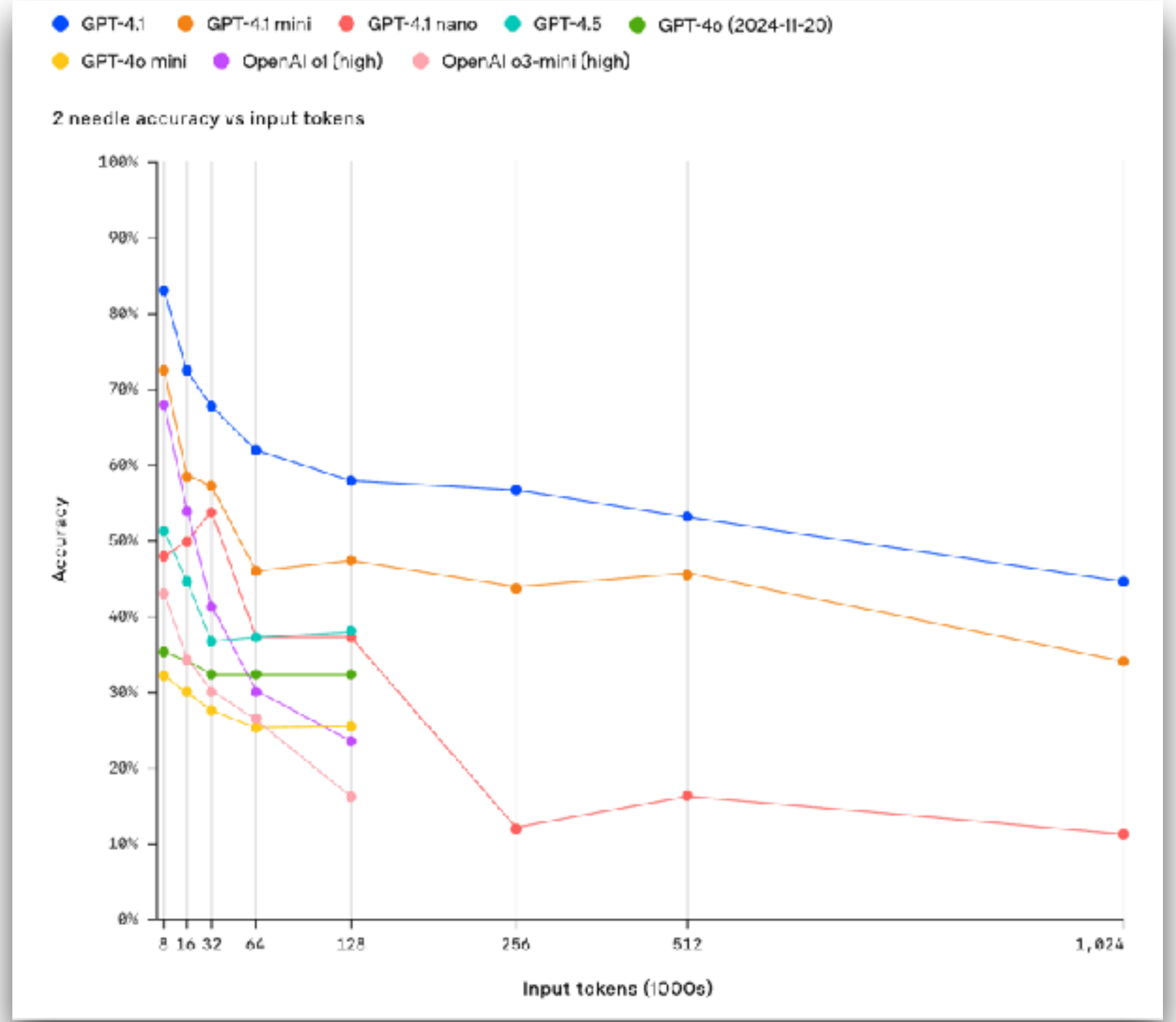
Long context LLM is better !!



https://openai.com/index/gpt-4-1/

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Better context == Better result ?

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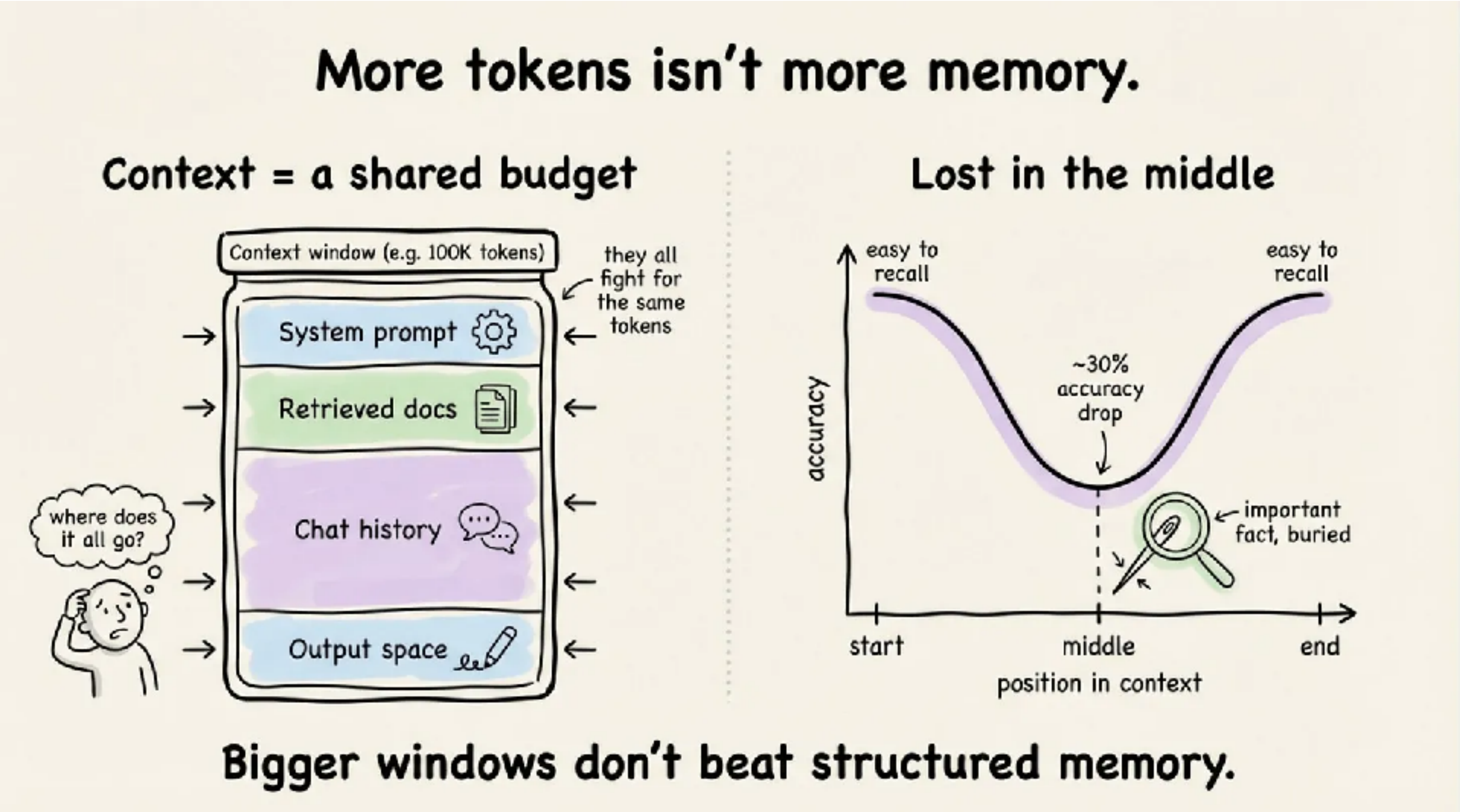

https://blog.dailydoseofds.com/p/evolution-of-agent-landscape-from

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Agent Memory
Memory Bank

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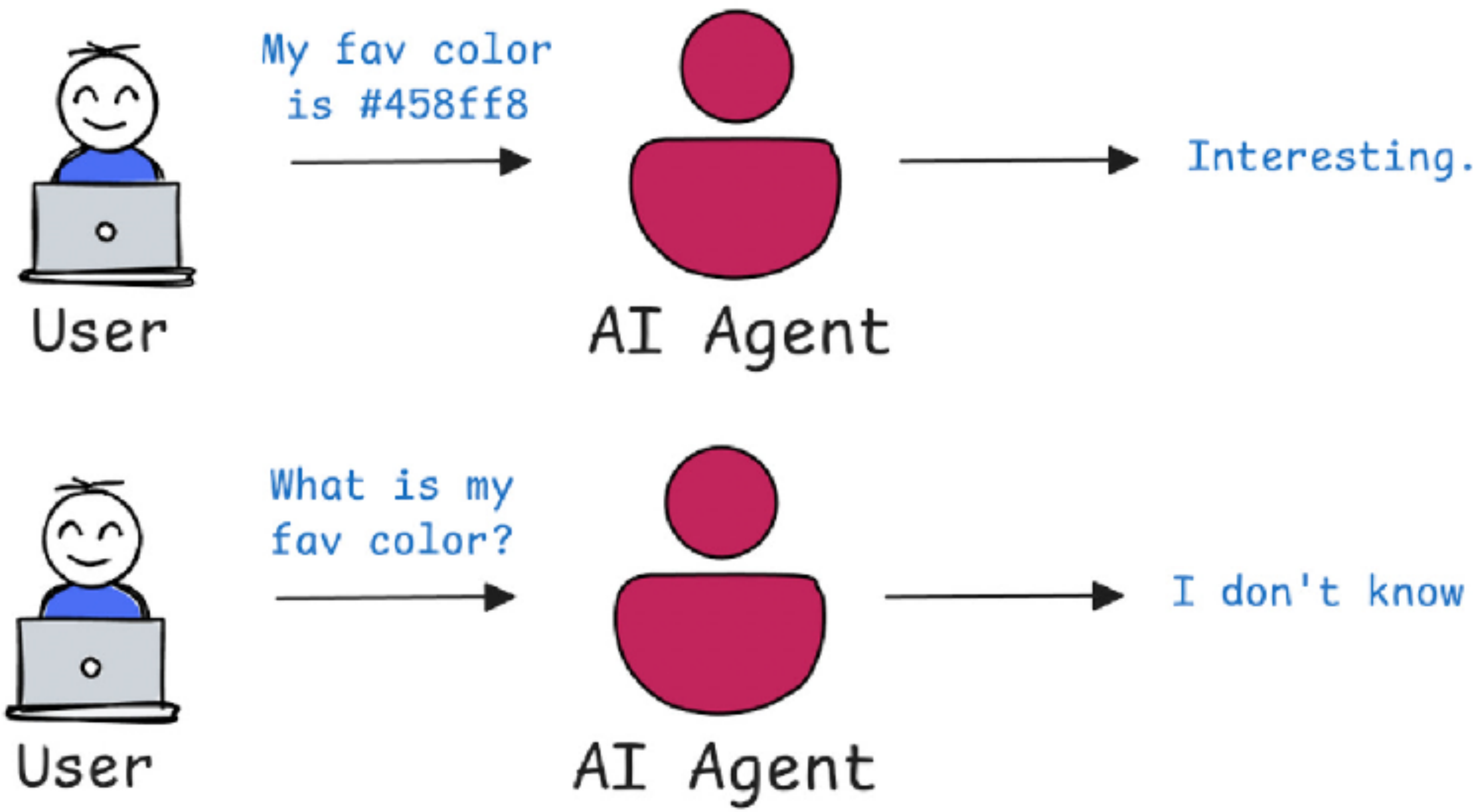
Interaction without memory



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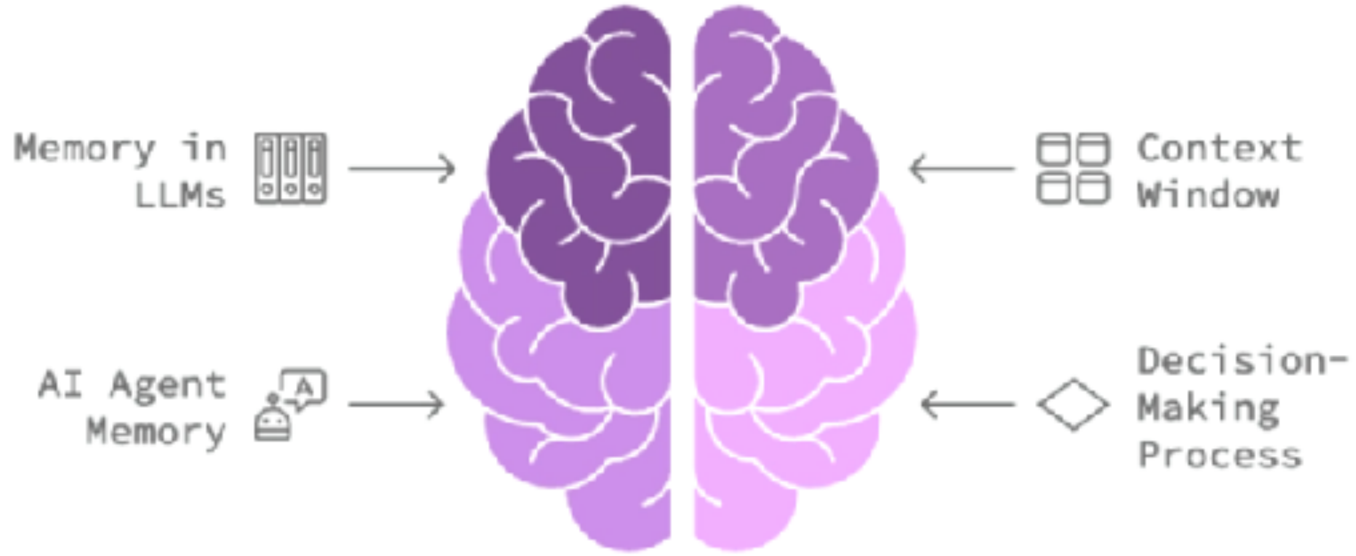
Why AI Agent memory important ?

Enable personalized and context-aware interactions
Improve decision-making and adaptability
Enhance LLM’s performance and response



https://www.falkordb.com/blog/ai-agents-memory-systems/

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Types of Memory

To make intelligent decisions
Agents combine different types of memory

Short-term

Current goals and steps
File system (markdown, JSON)
In-memory database

Episodic

Logs of past interactions
Success and failure

Semantic

Long-term knowledge
Fact and strategies
RAG
Vector and graph database

https://bhavishyapandit9.substack.com/p/how-memory-works-in-agentic-ai-a

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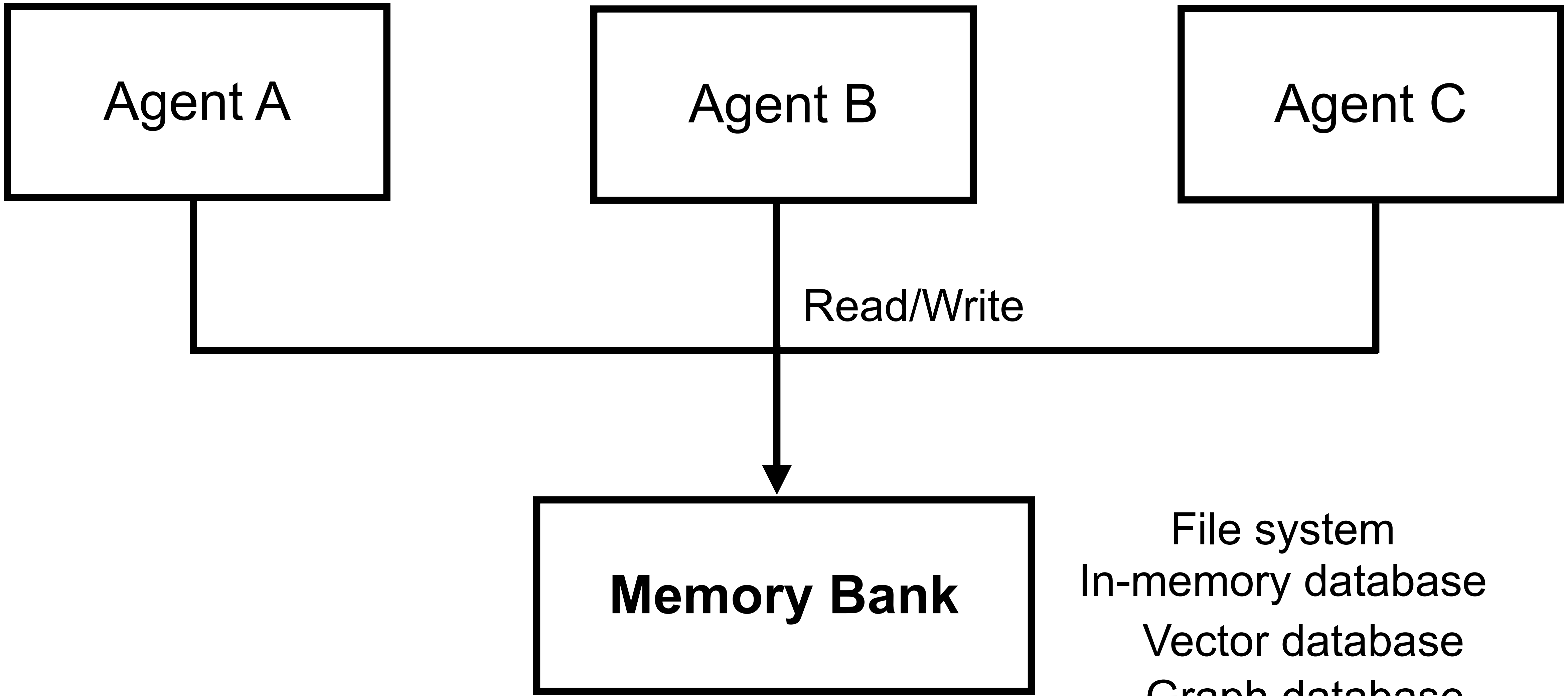
Shared memory between Agent



File system
In-memory database
Vector database
Graph database

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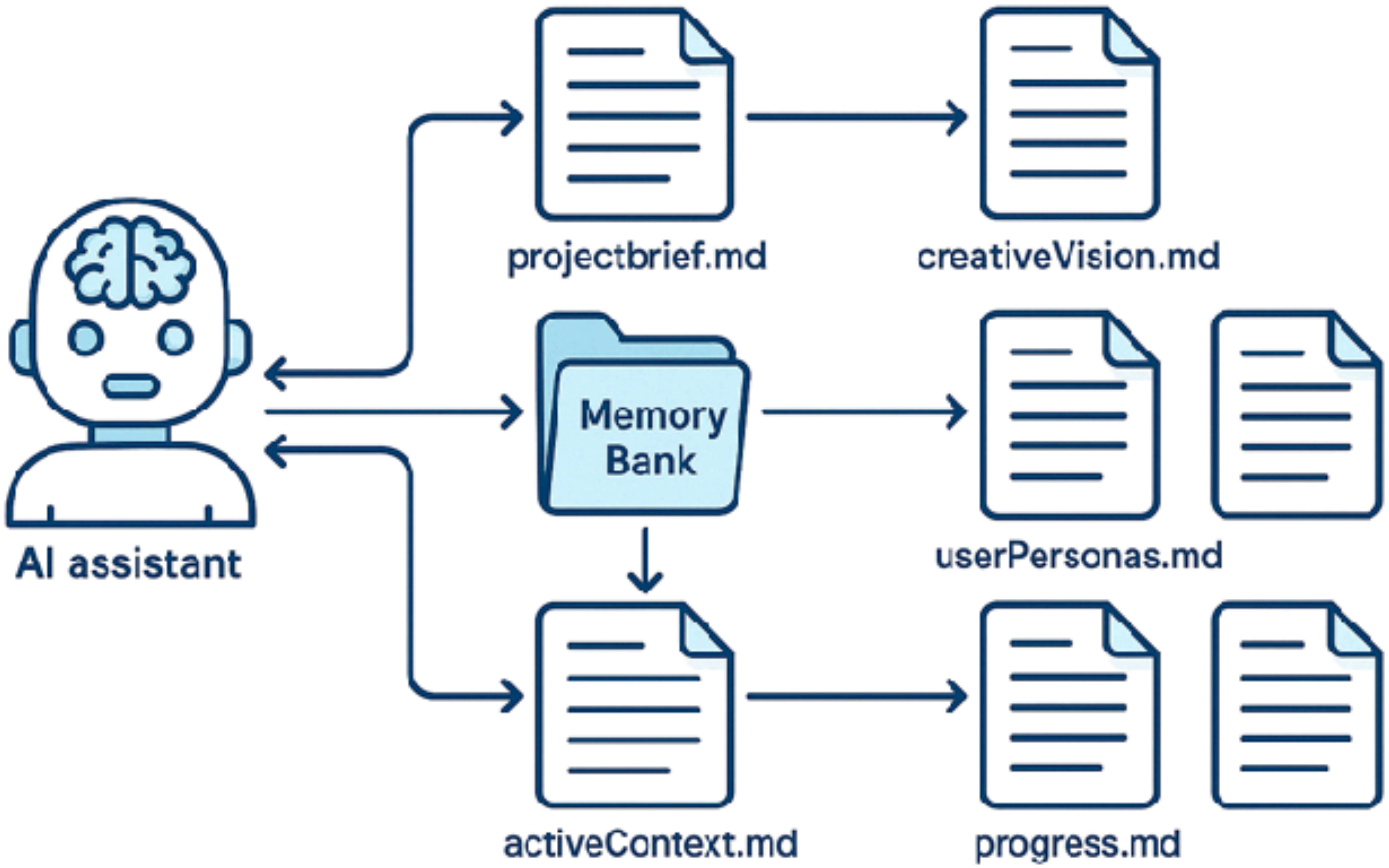
Cline Memory Bank

Working with file system



https://docs.cline.bot/prompting/cline-memory-bank

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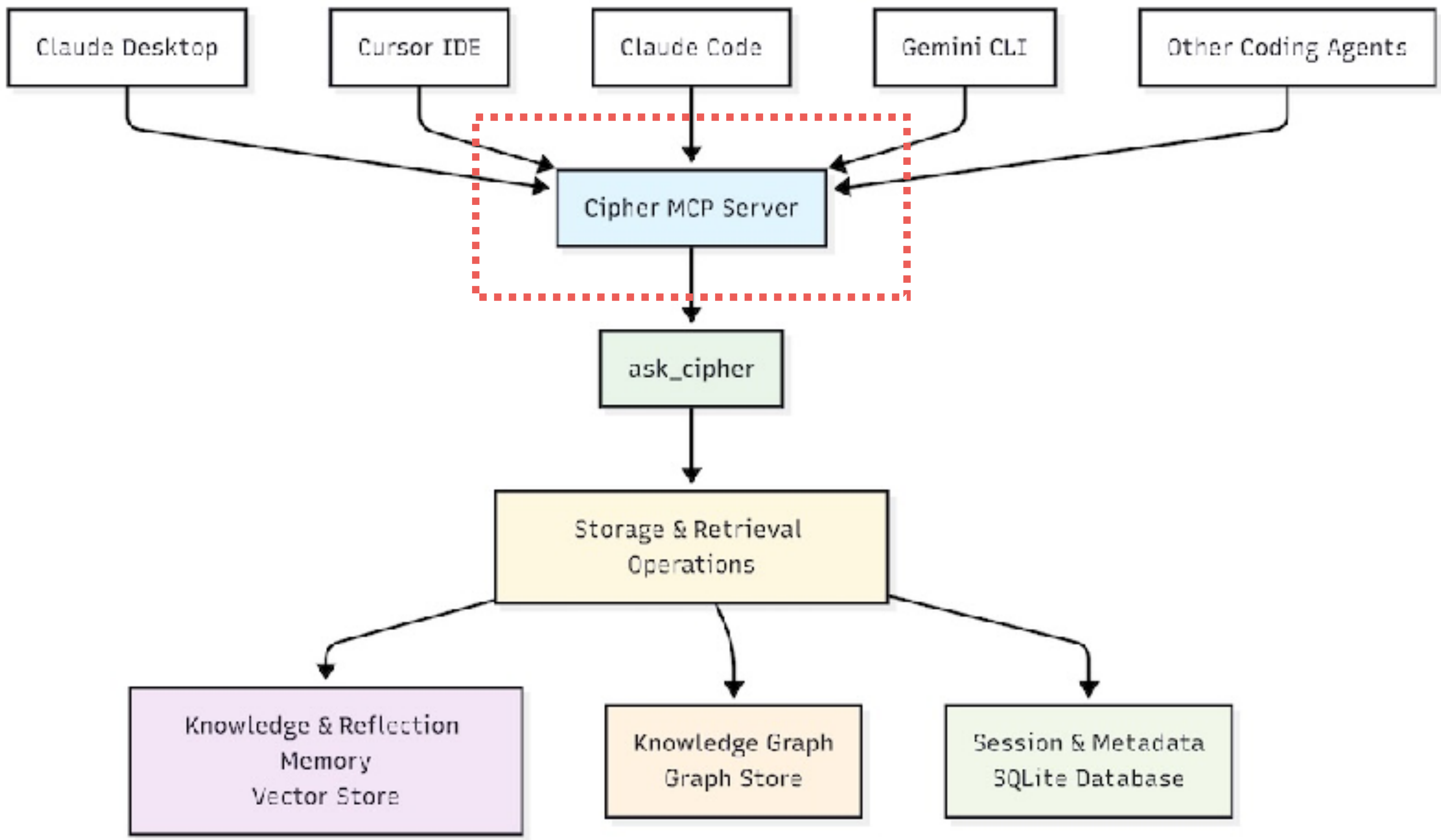
Memory Bank via MCP



https://www.somkiat.cc/share-memory-for-ai-ide/

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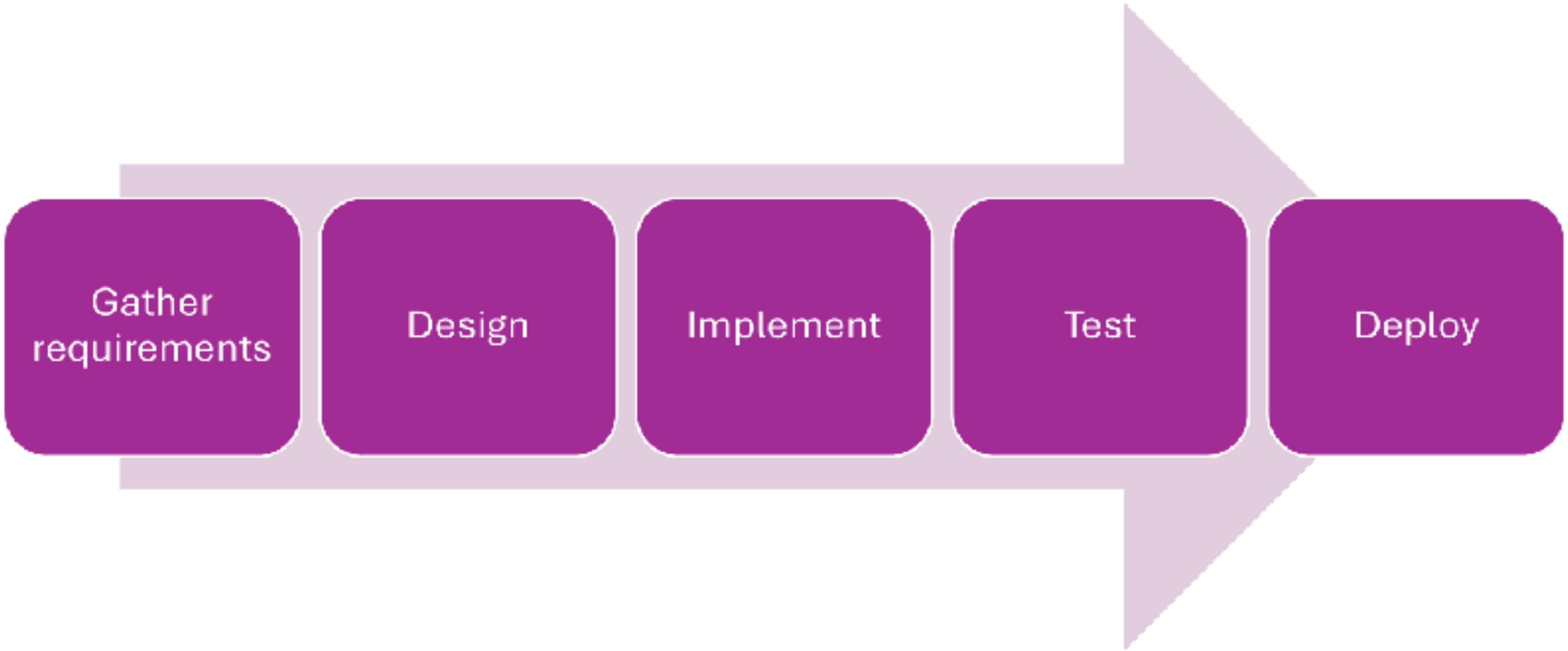
Design your AI Agents ?



https://devblogs.microsoft.com/dotnet/introducing-microsoft-agent-framework-preview/

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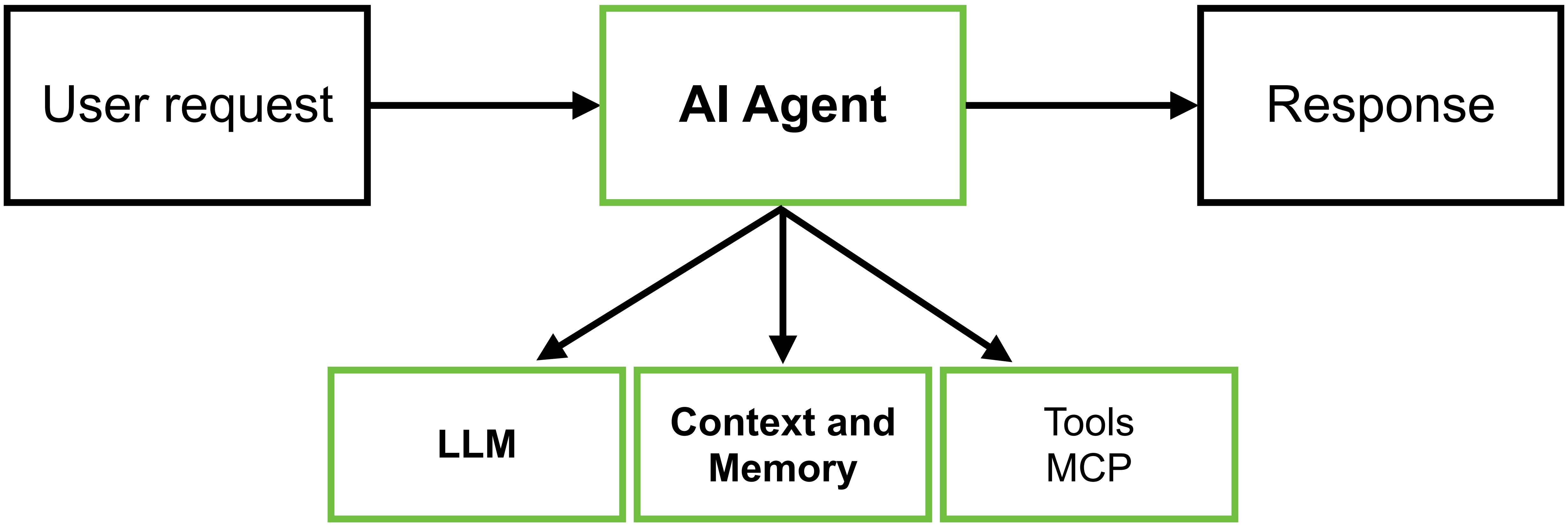
Single Agent



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Sub-agent and Multi-agent !! (Teamwork)

Reduce size of knowledge !!

Single Responsibility Principle

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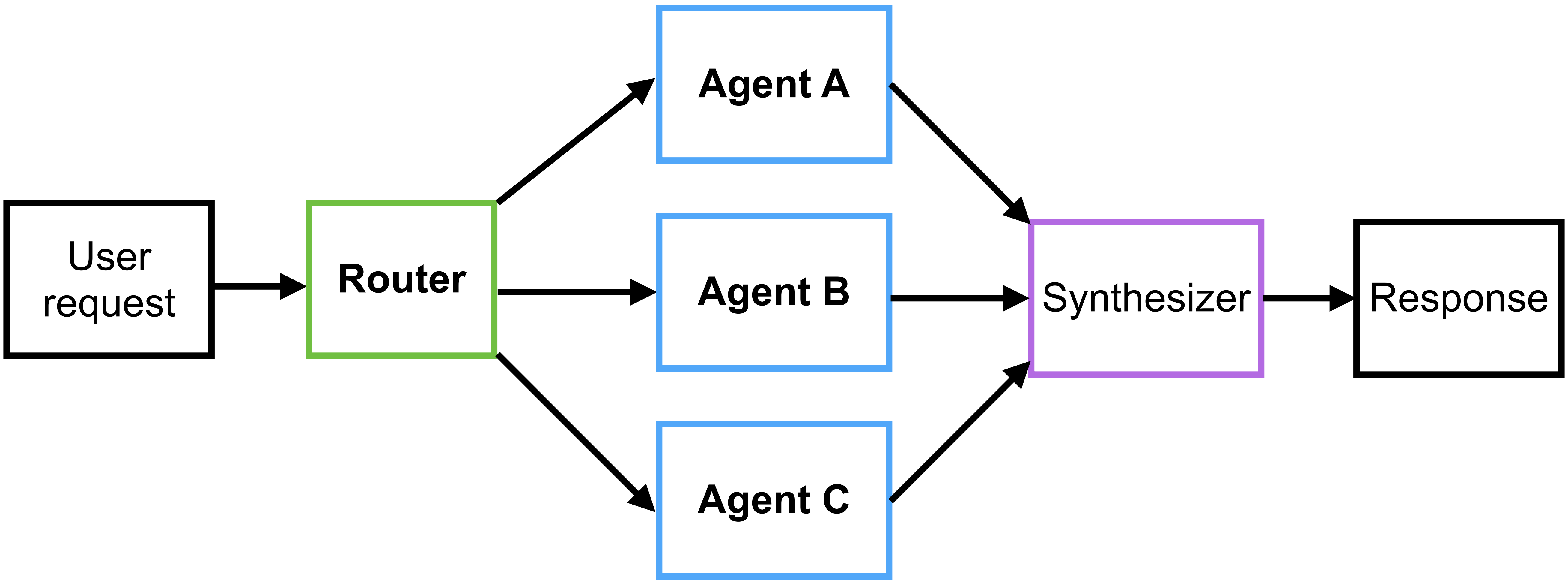
Multi AI Agents !!

Router and parallel run



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Manage context !!
In Software development

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Specification ?
Architecture ?
Modeling ?
Coding convention ?
Project structure ?
Better practice ?
Technology stack ?

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Model-Driven Development (MDD)

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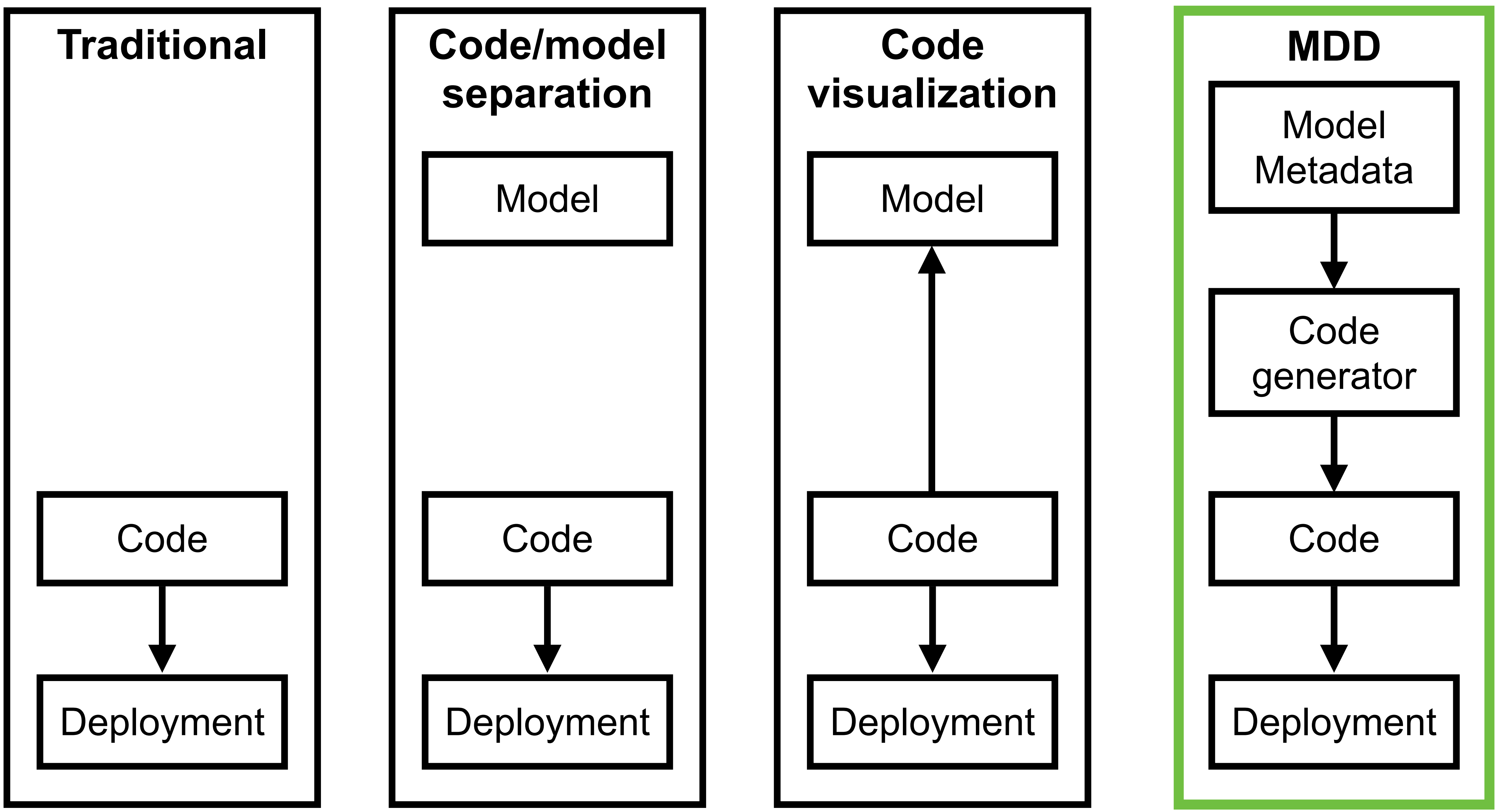
Traditional coding to MDD



https://arxiv.org/abs/2410.18489

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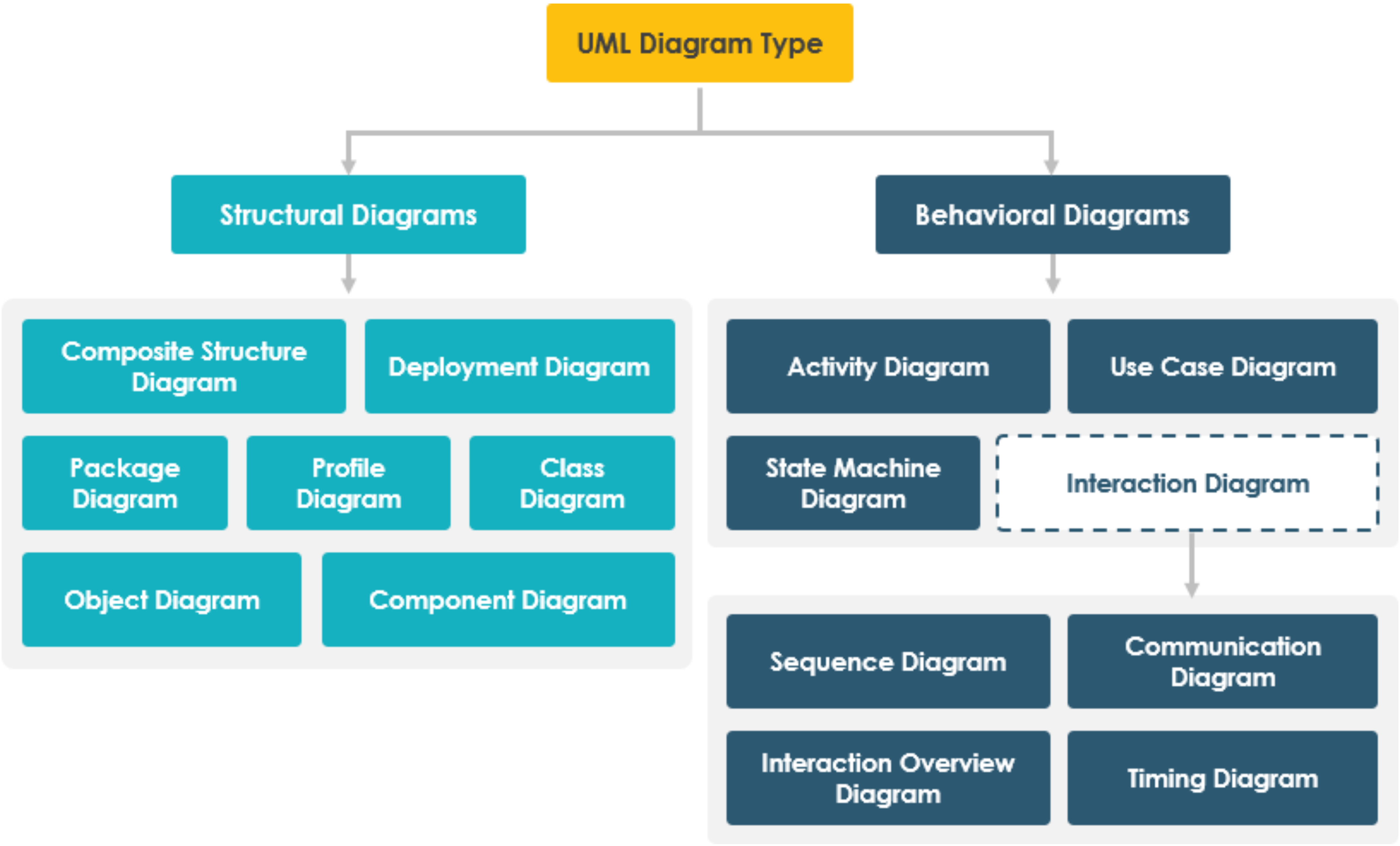
UML Diagrams !!



https://www.visual-paradigm.com/guide/uml-unified-modeling-language/behavior-vs-structural-diagram/

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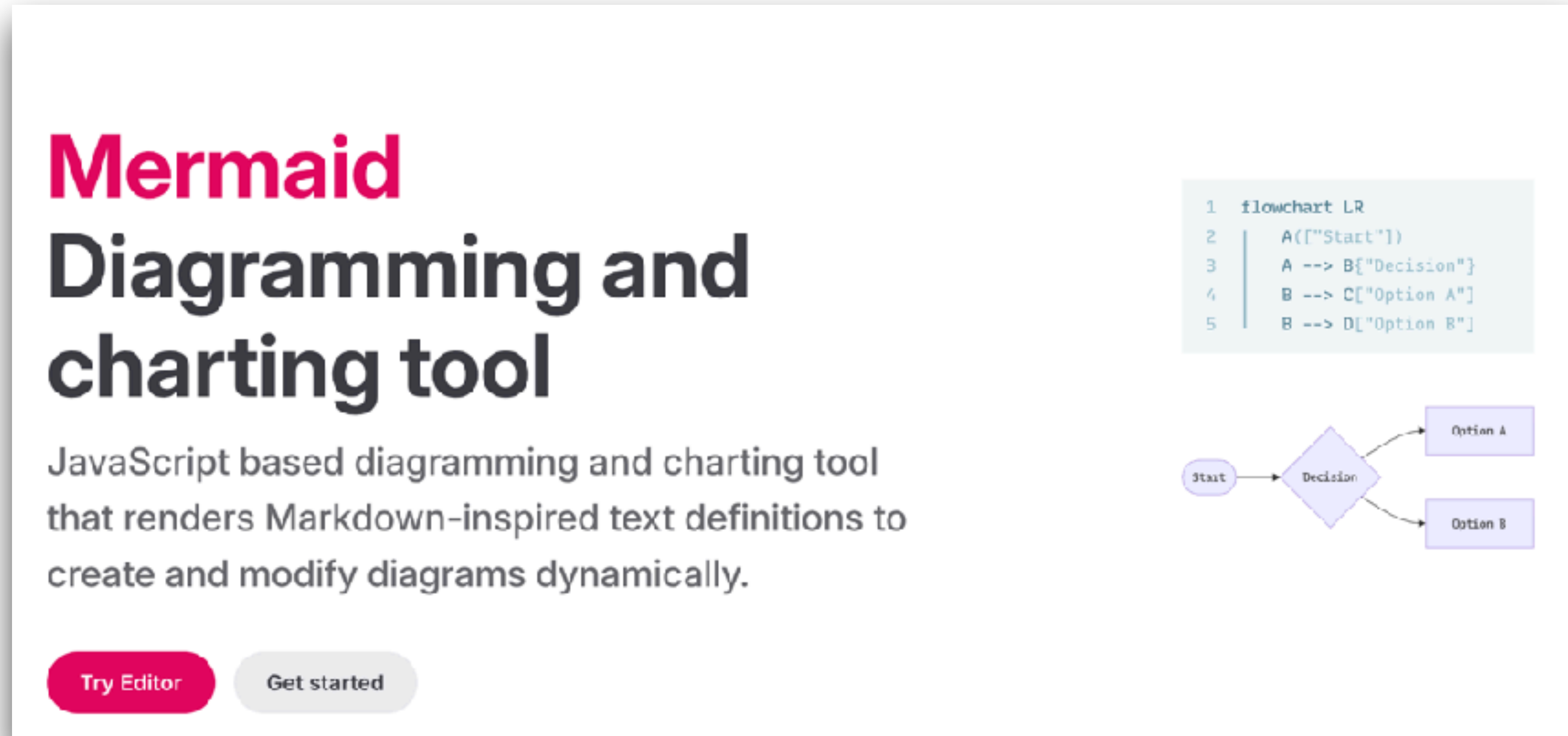
Mermaid Diagram

Diagram as a Code (AI-friendly)



https://mermaid.js.org/

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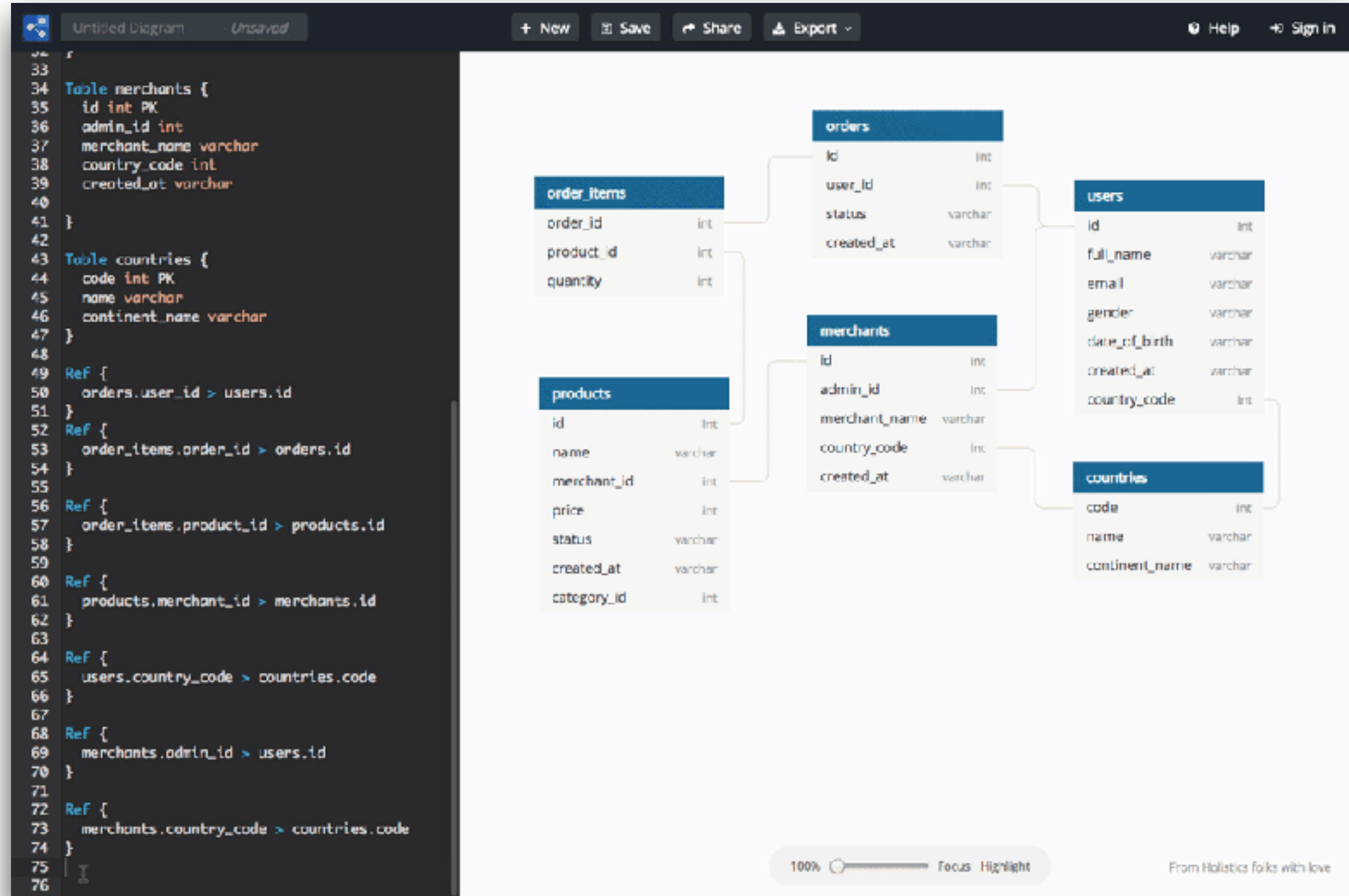
ER Diagrams for Database !!



https://dbml.dbdiagram.io/home/

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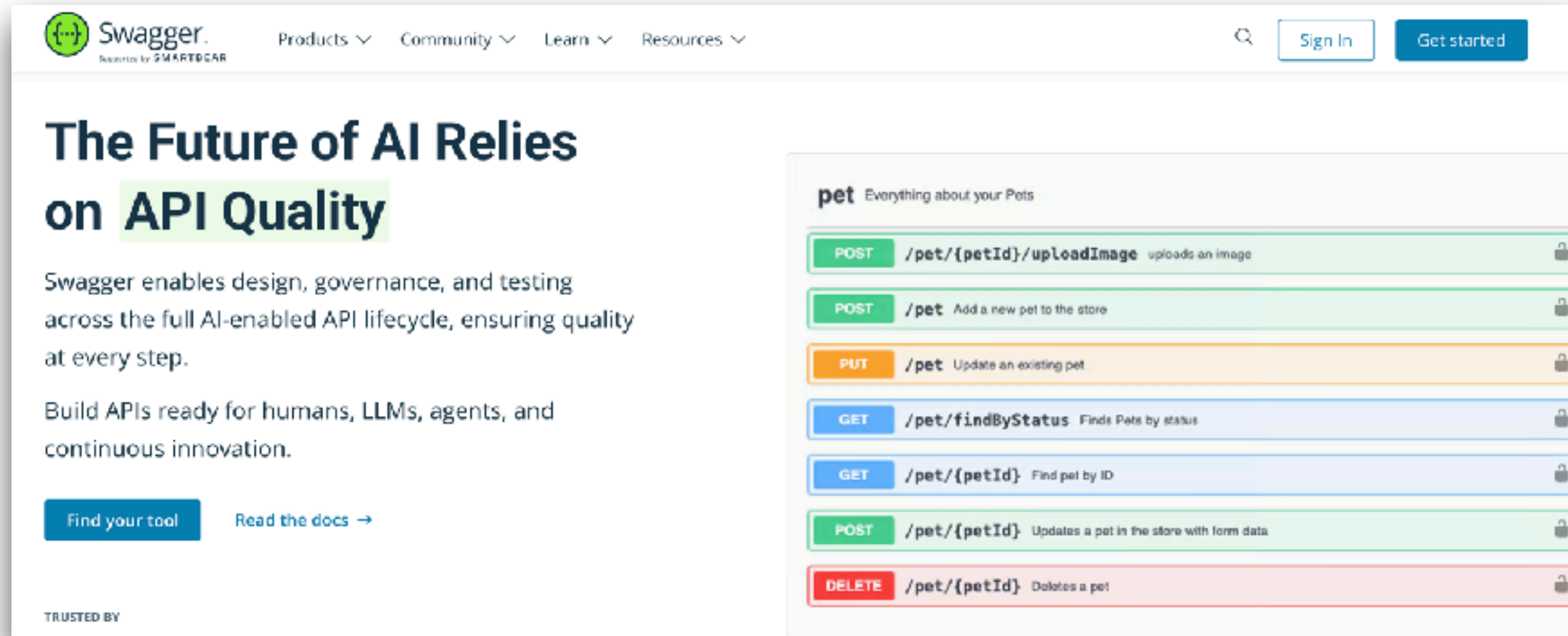
Swagger or OpenAPI

Standardized specification for describe REST APIs



https://swagger.io/

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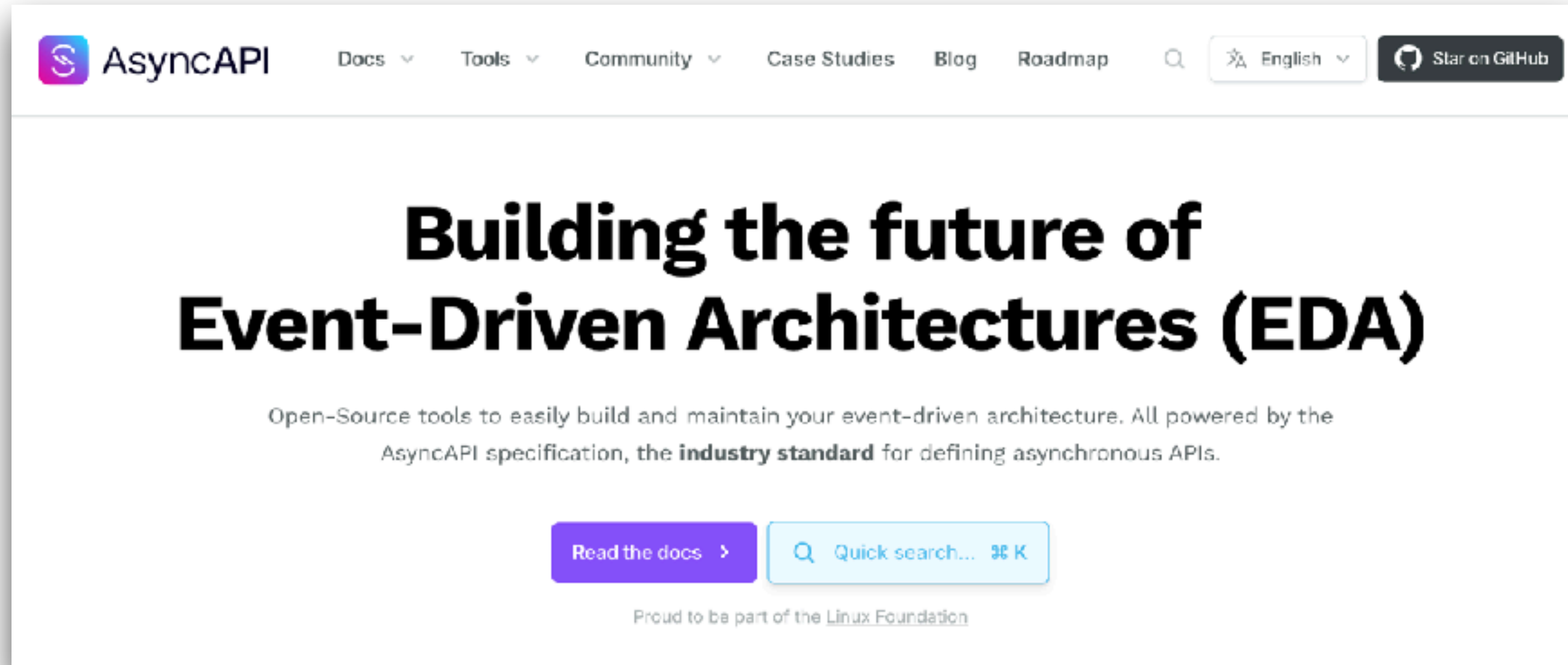
AsyncAPI

Standardized specification for describe asynchronous and event-driven APIs (Kafka, AMQP, WebSocket)



https://www.asyncapi.com/

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JSONSchema

Declarative language used to annotate and validate the structure, constraints and data type of JSON documents

Data validation

Documentation

Automated testing

https://json-schema.org/

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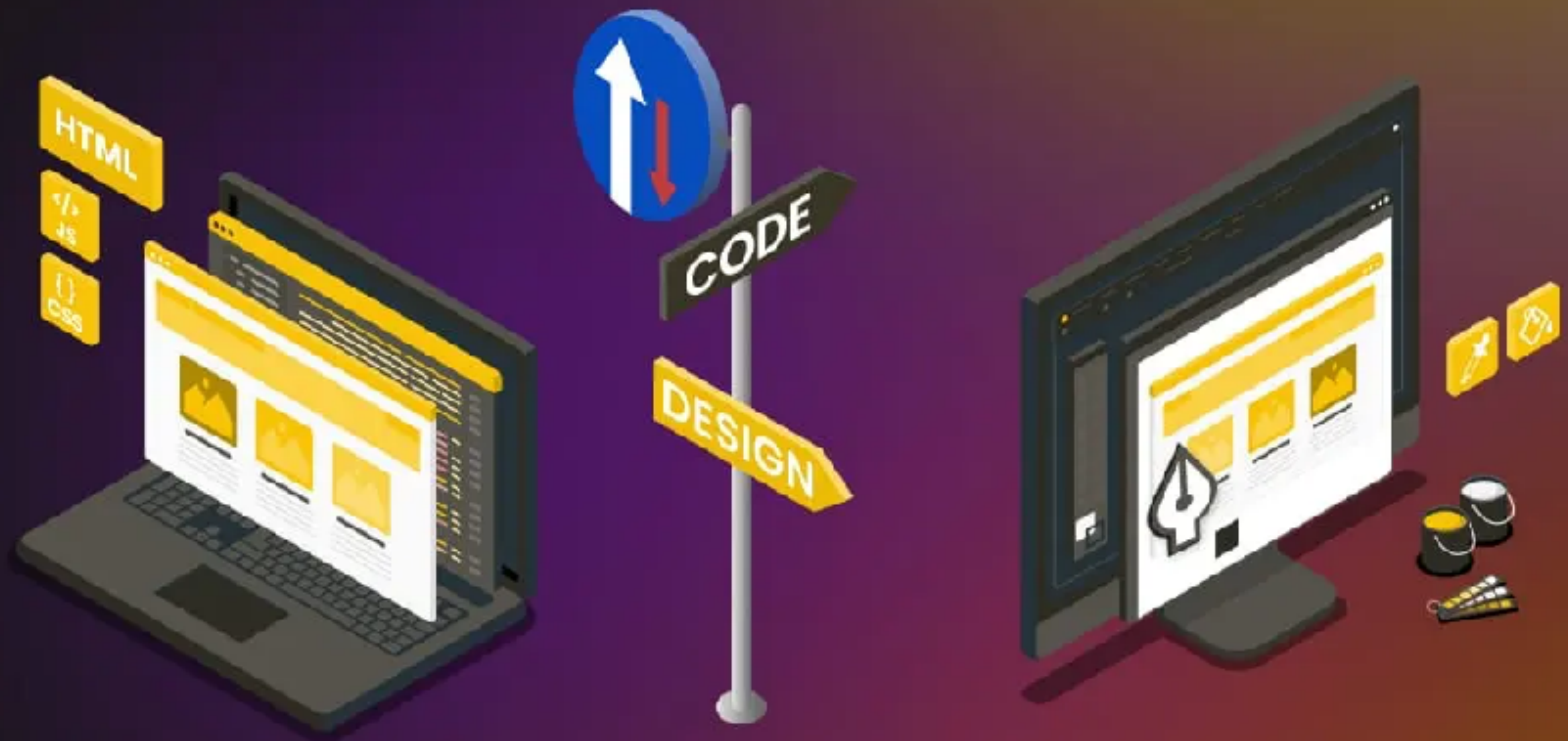
Design to Code
Code to Design



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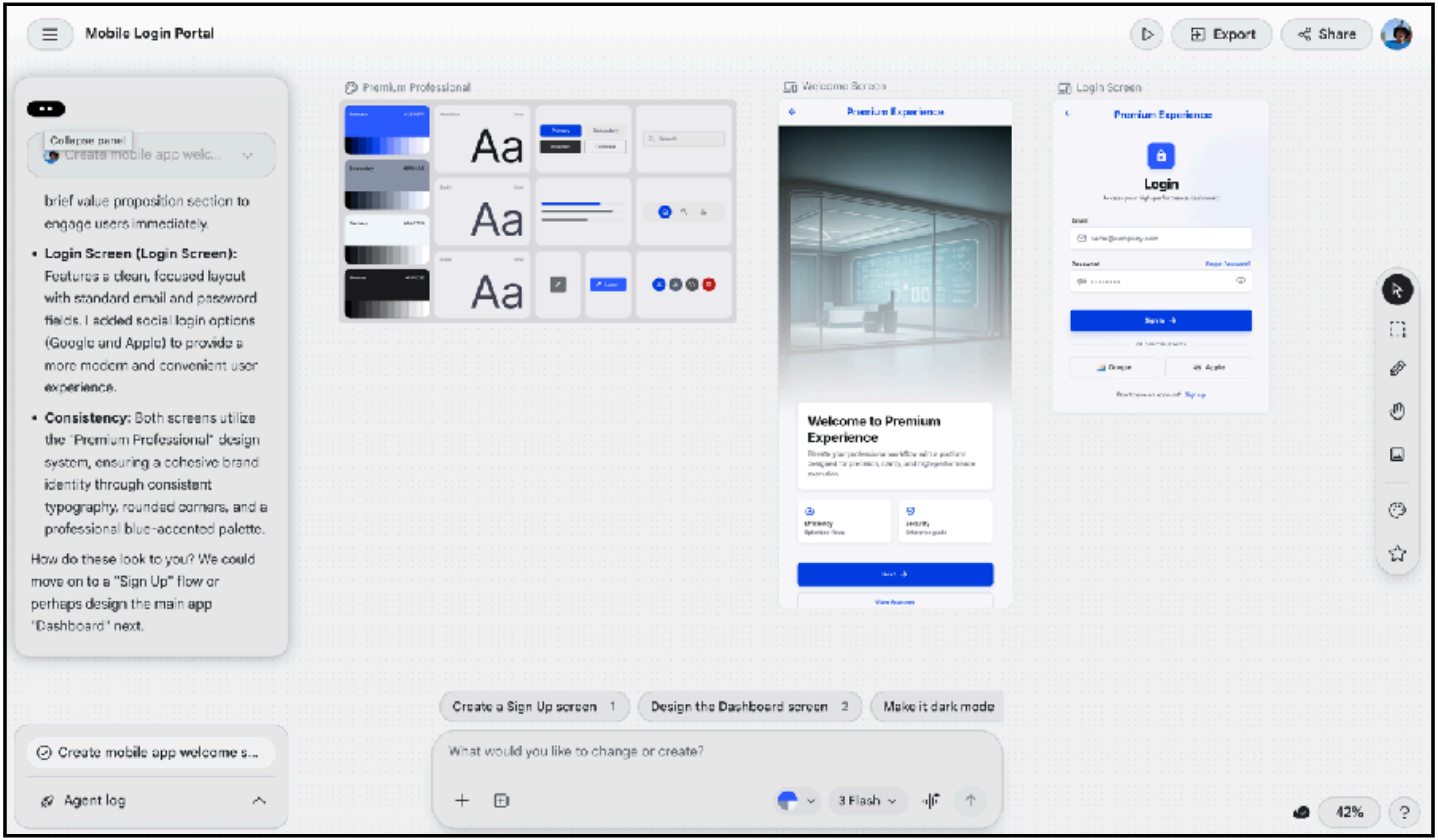
Google Stitch



https://stitch.withgoogle.com/

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Spec-Driven Development (SDD)

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SDD ?

Specification or Documentation-First

Modern software development approach where detailed, structure specifications are created before writing code

Idea → Requirement → Design → Tasks → Execute Coding

Validate and Decision log

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Types of Specification ?

User requirements
Design documents
Interface contracts (UI, API)
Data schemas
Validation rules
Data/business flows
Security policy and constraints
Resources and performance constraints

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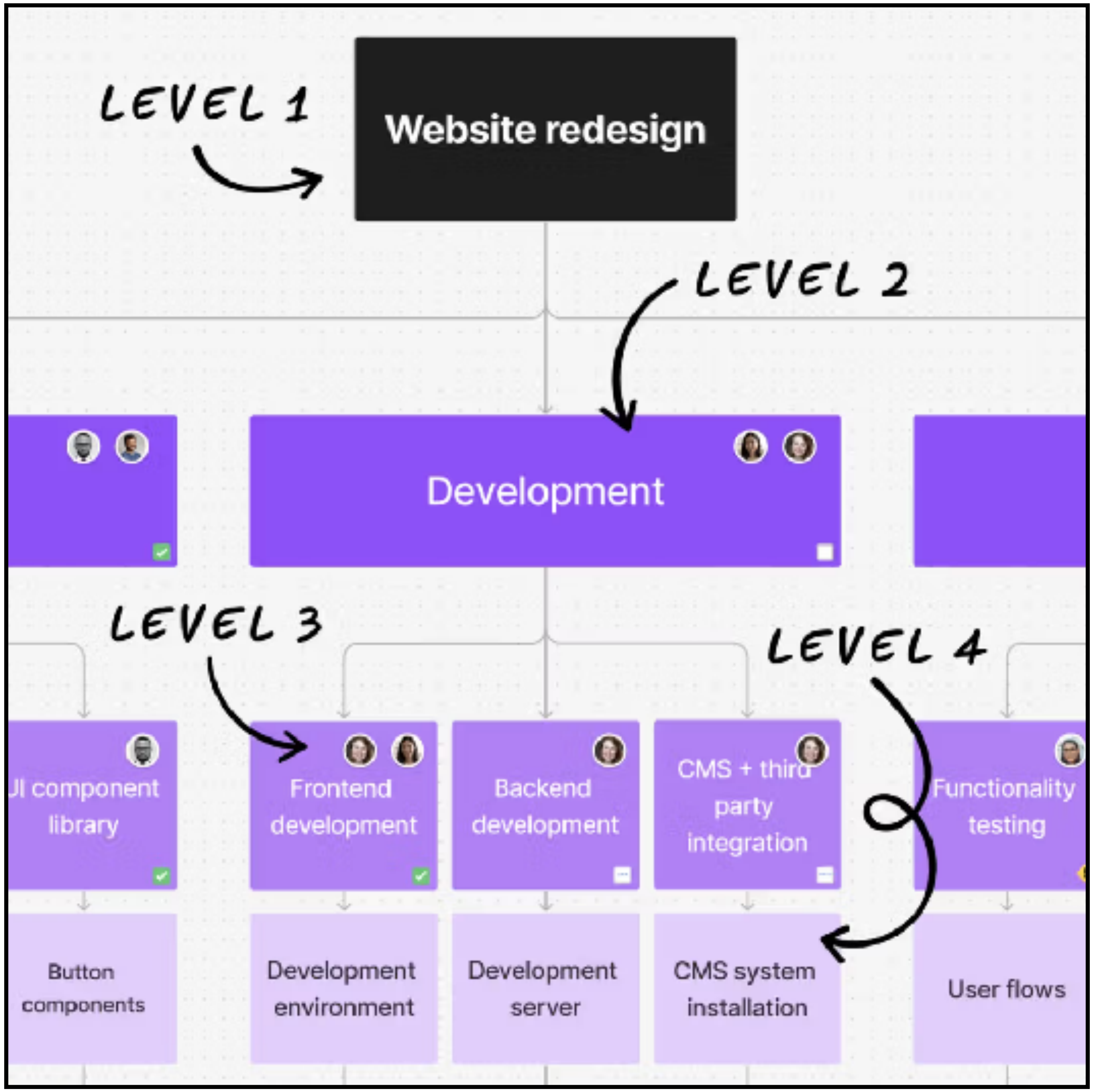
Work Break Down (WBS)



https://www.figma.com/resource-library/work-breakdown-structure/

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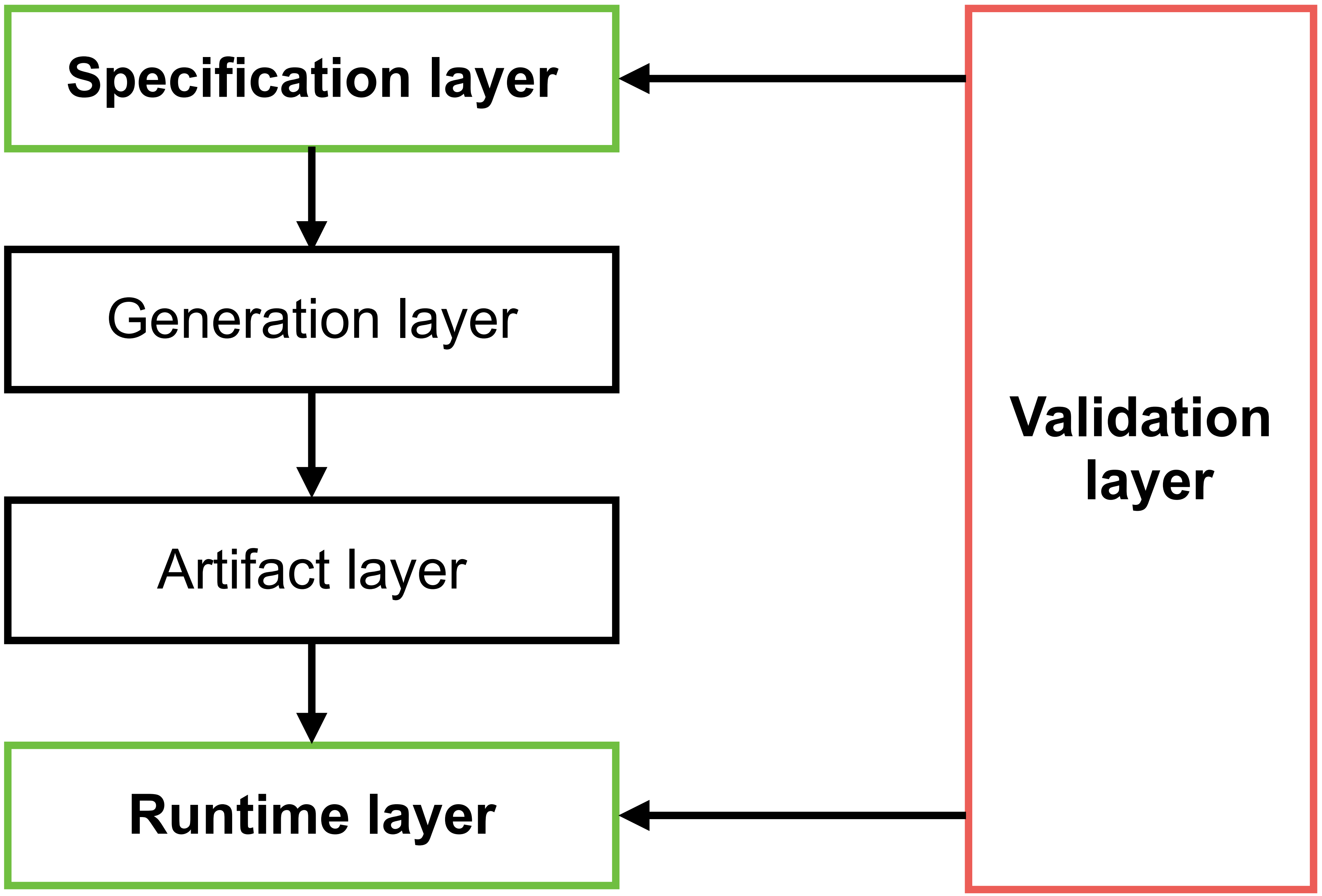
SSD Architecture



https://www.infoq.com/articles/spec-driven-development

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MDD + SDD == Better result

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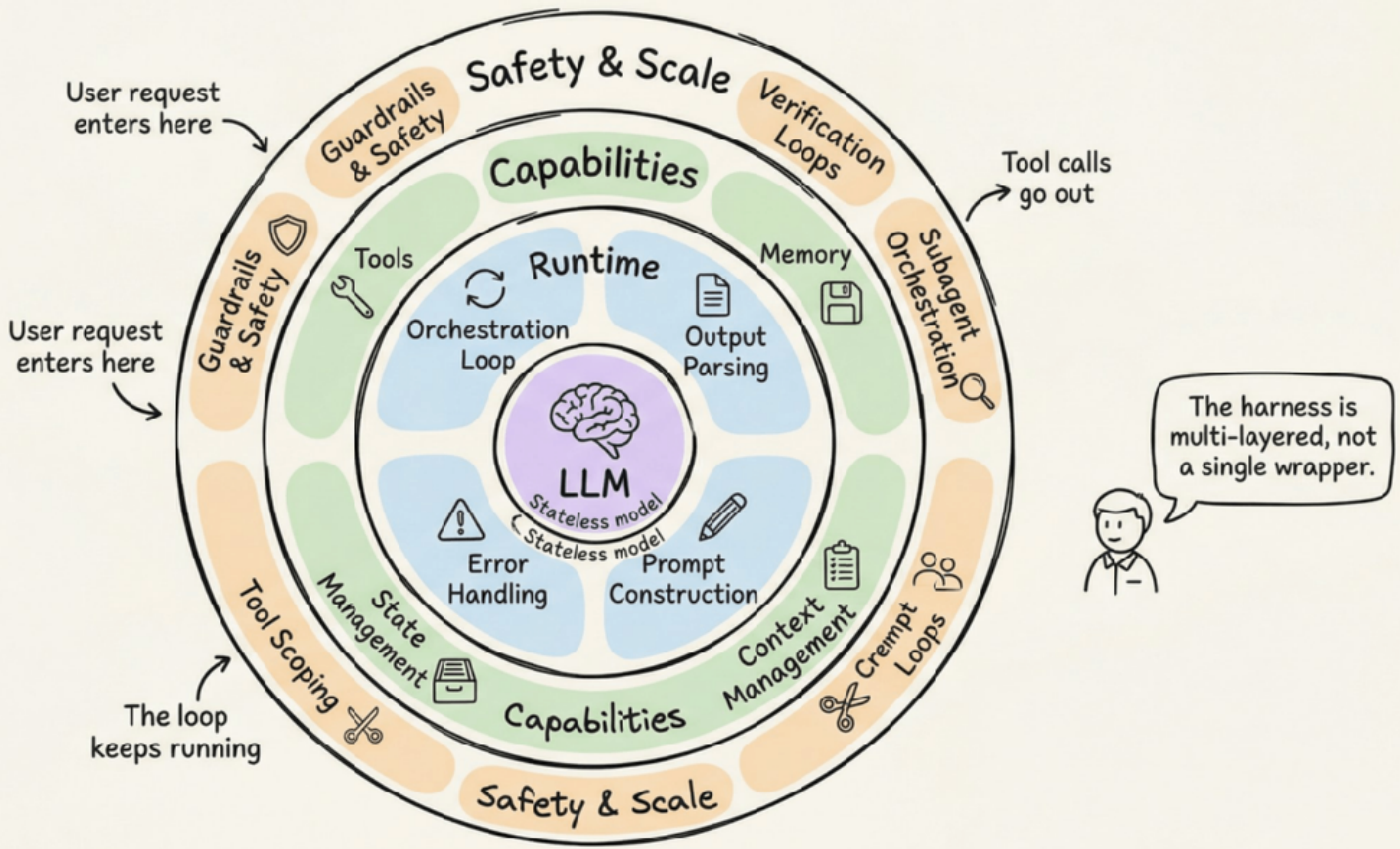

https://blog.dailydoseofds.com/p/evolution-of-agent-landscape-from

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Demo time

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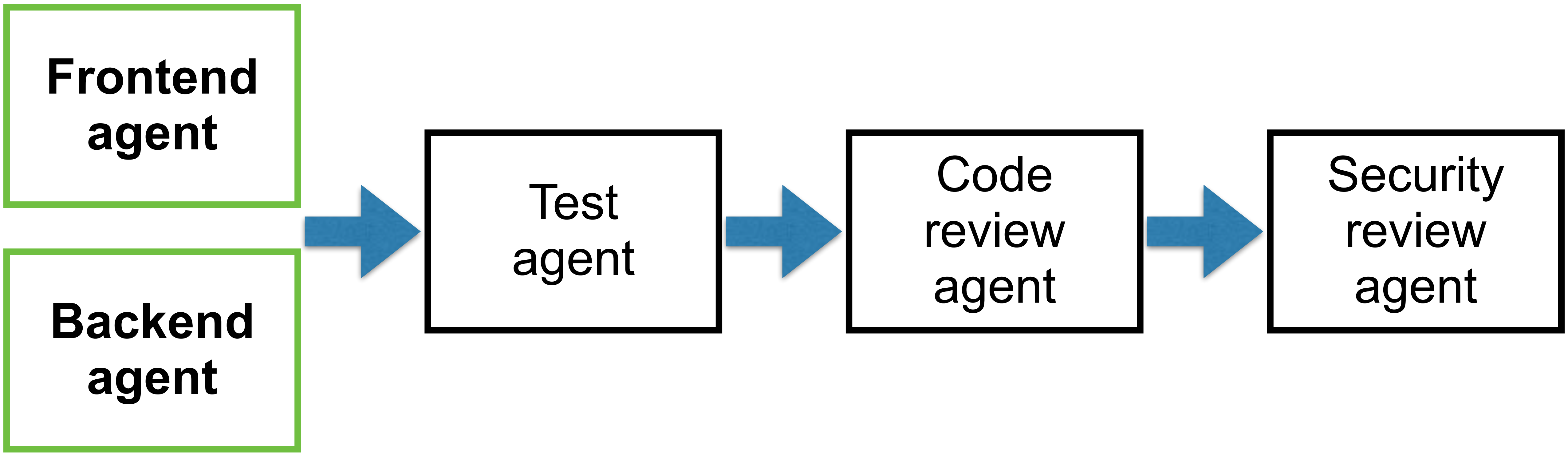
Design your Agents



Manage memory and knowledge of project !!

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Frontend
Mobile application

Backend
RESTful API + Database

Write specification

Coding with AI Agent

Run, Test and Verify !!

Version management system

https://github.com/up1/demo-sharing-ai-agent-202605

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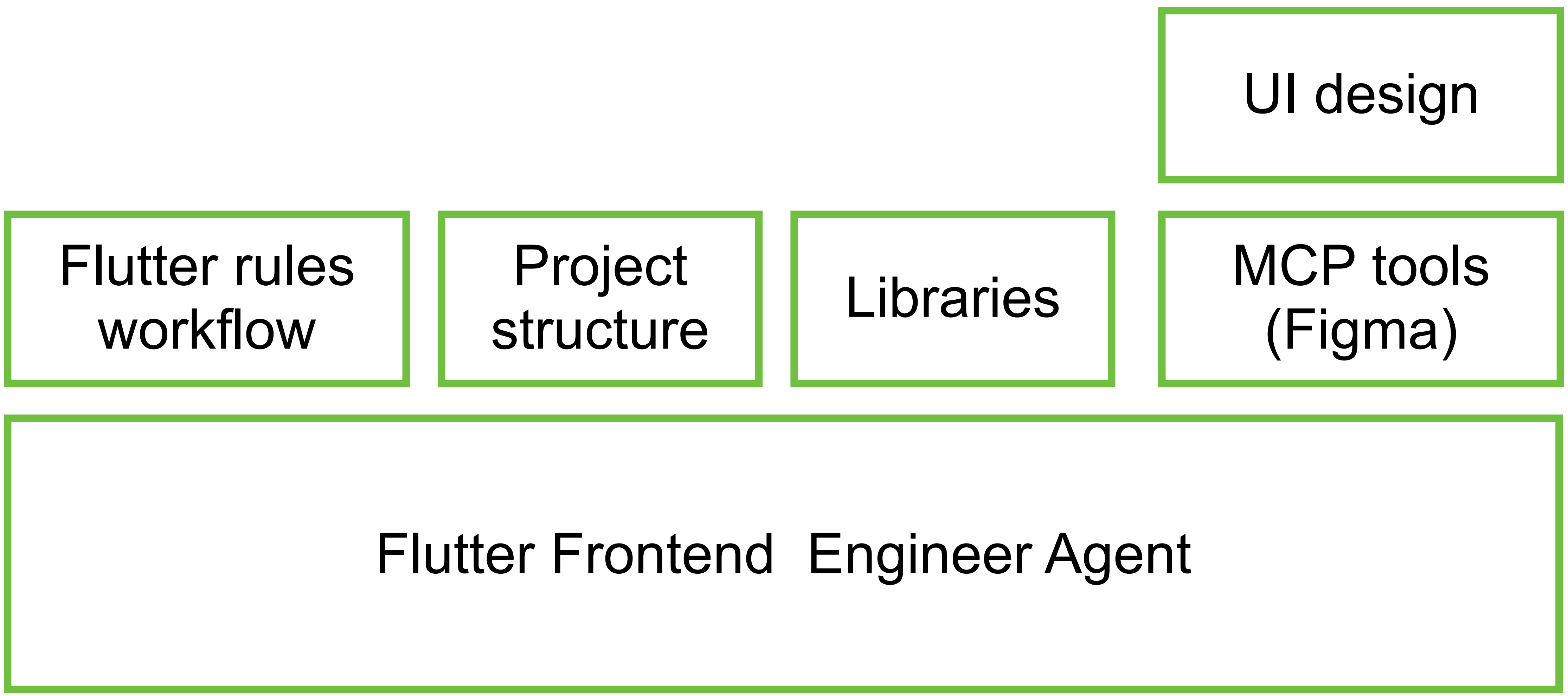
Frontend Agent (1)



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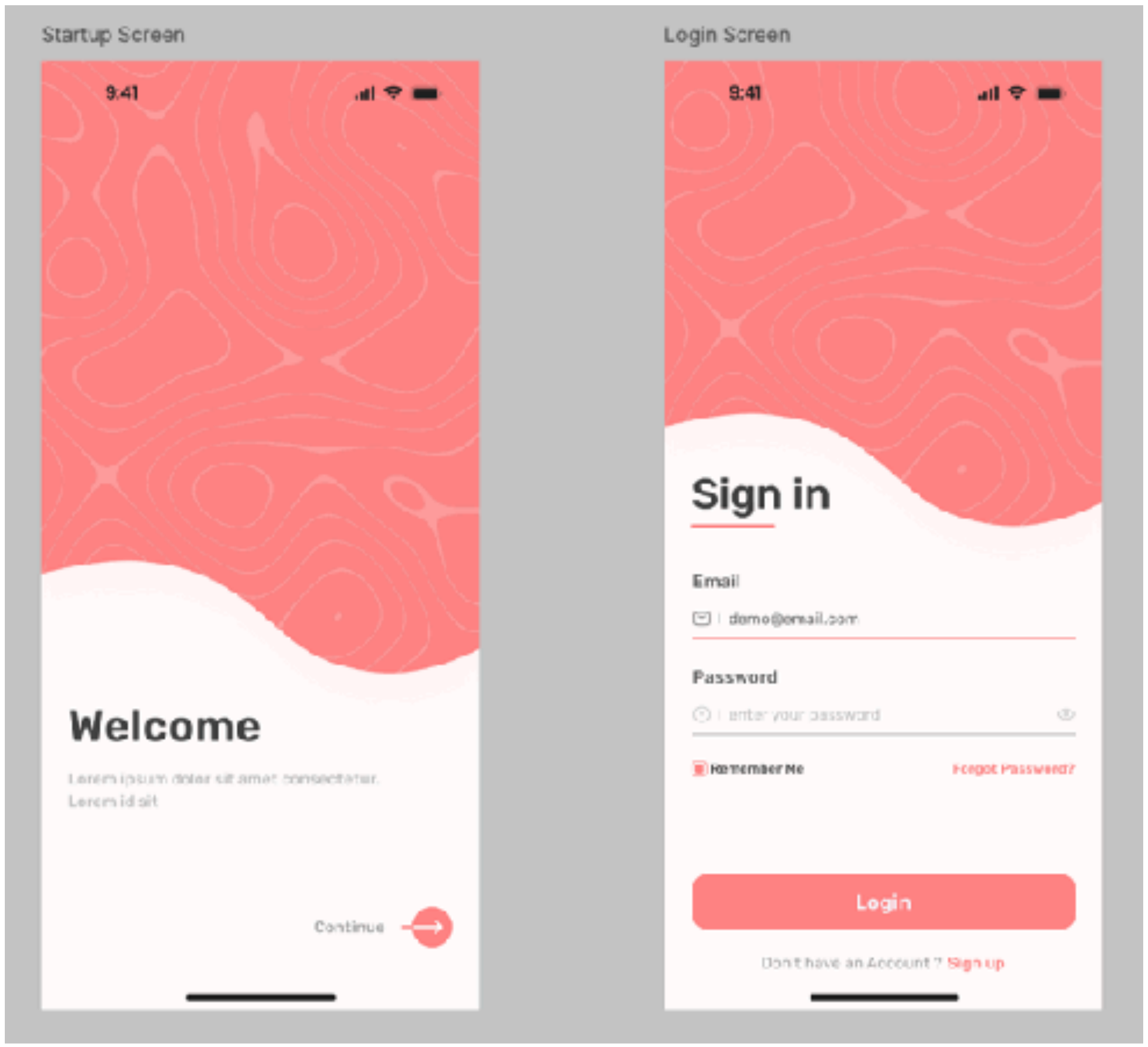
User Interface Design

Design to Code technique



Export → HTML/CSS

https://bit.ly/3P35iNU

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Frontend Agent (2)

Write specification

UI design

Flutter rules workflow

Project structure

Libraries

MCP tools (Figma)

Flutter Frontend Engineer Agent

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Backend Agent

Write specification

NodeJS rules workflow

Project structure

Libraries

NodeJS Backend Engineer Agent

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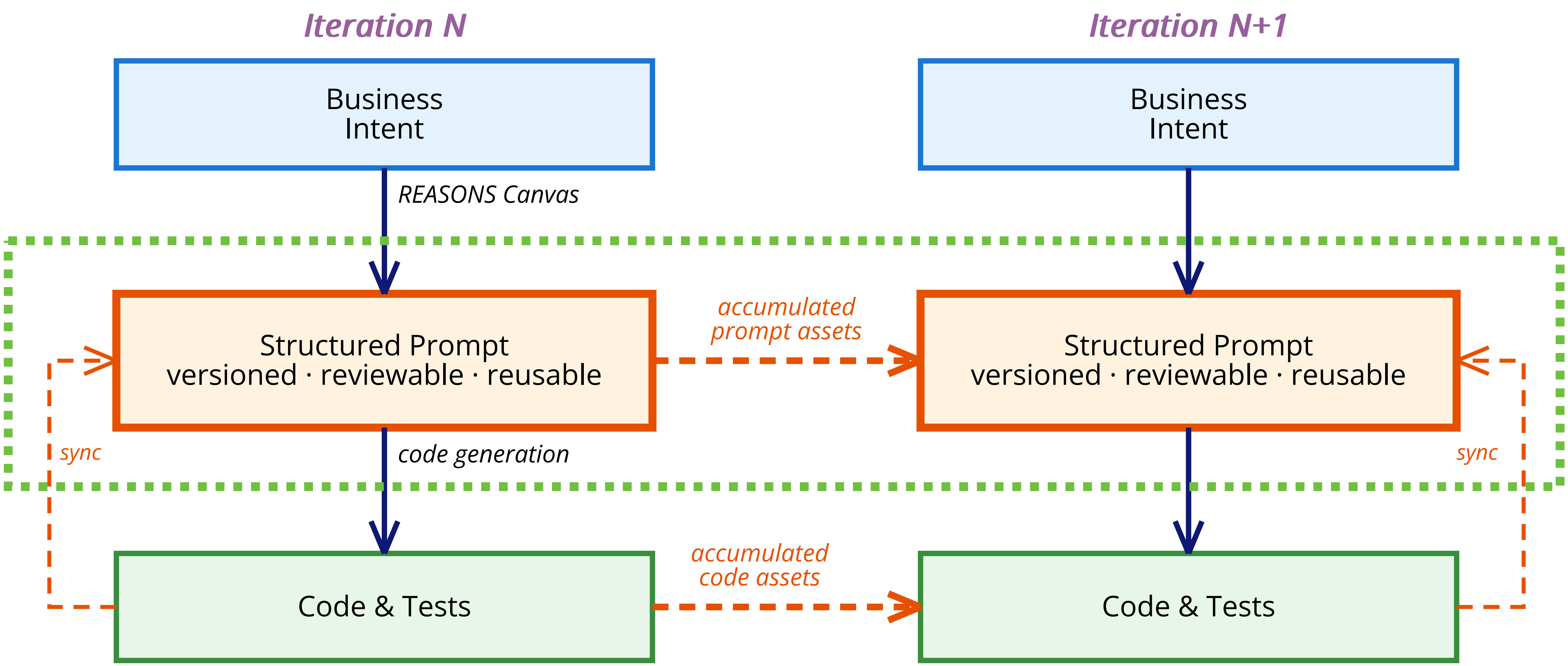
Structure of Specification

Reliable, Repeatable and Versioning



https://martinfowler.com/articles/structured-prompt-driven/

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Developer
Survival Checklists

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Developer Survival Checklists

1. Focus on logic (good or bad)
2. Audit AI Output (functional and non-functional)
3. Continuous Learning

Focus on foundation knowledge
to effectively work with and guide AI system

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AI draft
Human craft

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Slide and Demo

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